

Memory and Beliefs in Financial Markets: A Machine Learning Approach[†]

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Abstract

This paper explores the role of memory in shaping belief formation of financial market participants. We estimate a structural machine learning model of memory-based belief formation for a representative stock analyst using consensus earnings forecast revisions. The estimated model reveals significant recall distortions compared to a benchmark model trained to fit realized earnings revisions. Specifically, analysts over-recall distant historical episodes during normal times, when recent episodes are more useful for forming forecasts than those in the distant past, but under-recall them during crisis times, when history helps interpret unusual episodes. These distortions arise partly from overweighting past earnings and forecasts and from how information is selectively retained in memory. Our estimated analyst recalls predict earnings forecast revisions, stock returns, and market disagreement, while the recall distortions predict forecast errors and asset mispricing. (*JEL codes*: D91, G41, G12, G14)

Keywords: Memory, Machine Learning, Belief Formation, Mispricing, Analyst

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1 Introduction

“In response to current events, people often reach for historical analogies, and this occasion was no exception. The trick is to choose the right analogy.” - [Bernanke \(2015\)](#)

Belief formation is central to asset pricing, yet we know little about how people actually form beliefs in financial markets. Research in psychology highlights the role of memory in belief formation: people draw on past episodes to interpret new information and update their beliefs ([Kahana, 2012](#)). Recent studies in finance and economics have introduced memory-based theories of belief formation ([Mullainathan, 2002](#); [Bordalo et al., 2020](#); [Wachter and Kahana, 2024](#)) and provided supporting evidence from surveys and laboratory experiments ([Bordalo et al., 2025](#); [Jiang et al., 2025](#); [Enke et al., 2024](#); [Gödker et al., 2025](#); [Graeber et al., 2024](#)). However, field evidence from financial market data on how memory influences belief formation and asset prices remains limited.¹

Studying memory-based belief formation in the field is challenging because market participants’ memory is not directly observable. We address this challenge by developing a new empirical approach that infers memory from consensus analyst earnings forecasts. The method recovers the memory of a representative analyst for each firm, which we interpret as the market’s collective memory for that firm. Within this framework, we provide novel field evidence by examining two fundamental components of memory: mental context and recall. Mental context represents how the analyst perceives the firm’s current state, summarizing the most relevant information and guiding the interpretation of new signals. Recall captures which past experiences the analyst draws on when forming beliefs, identifying past episodes (any historical observation of a firm) that resemble the current one based on the similarity of their mental contexts.

Our empirical approach involves two steps. In the first step, we estimate analyst mental context using a structural memory model – Long Short-Term Memory (LSTM). It is a neural network that closely resembles the state-of-the-art memory model in psychology and

¹[Jiang et al. \(2025\)](#) combine survey and transaction data to show that memory affects investors’ trading. [Charles \(2025\)](#) study how memory-based attention impacts stock prices around earnings announcements.

economics by Wachter and Kahana (2024).² LSTM contains a dynamic memory cell that retains past information and updates it as new signals arrive, weighting memories by their relevance to current conditions. In each period, the representative analyst first observes high-dimensional external features, which are public signals including macroeconomic variables and firm fundamentals. Next, she updates her mental context by summarizing the most relevant information from the memory cell and new features to describe the firm’s current state. This mental context then guides analyst earnings forecast. We estimate the model using monthly data on consensus one-year-ahead earnings per share (EPS) forecast revisions and public signals for about 1,300 U.S. firms from 1990 to 2020, which allows us to infer the latent mental context for each firm in each month.

In the second step, we extract analyst recalls cued by the estimated mental context. When forming recalls for the current firm, the analyst compares its current mental context with those from past episodes across firms in the same industry. For each past episode, we compute the probability of recall as an increasing function of the similarity between its mental context and the current one (Kahana, 2012). This procedure generates a firm-month panel of inferred analyst recalls. Conceptually, current beliefs are associated with a similarity-weighted aggregation of recalled episodes (Bordalo et al., 2025). We then test whether these recall patterns predict beliefs and asset pricing outcomes.

Our empirical approach offers three advantages. First, LSTM provides a psychologically grounded model of human memory. Similar to the retrieved-context memory model of Wachter and Kahana (2024), it reproduces and allows us to empirically illustrate the well-established principles of human memory (Kahana, 2012): recency, temporal contiguity, and semantic similarity.³ Second, financial market participants often face high-dimensional, non-stationary variables with non-linear interactions. LSTM, as a machine learning model, is designed for such challenges, making it a natural tool to bring existing theoretical memory models (Bordalo et al., 2020; Wachter and Kahana, 2024) to empirical data. Third, our ap-

²LSTM was originally developed by Hochreiter and Schmidhuber (1997) in computer science. It has proven effective in modeling sequential dependence in language and time-series data, also paved the way for more advanced generative AI models.

³Recency: people tend to refer to recently experienced episodes when retrieving memories; temporal contiguity: people recall episodes that occurred close in time to the recalled episode; semantic similarity: people access episodes most similar to their current experiences.

proach can estimate analyst recalls for each firm over decades, providing large cross-sectional samples that are crucial for empirical asset pricing studies. This complements surveys and experiments, which typically focus on more specific settings (e.g., [Bordalo et al., 2025](#); [Enke et al., 2024](#)).

We provide empirical evidence on how memory shapes belief formation in financial markets along four dimensions: the extracted analyst recalls, the distortions in these recalls compared to a benchmark memory model, the underlying mechanisms behind these distortions, and their asset pricing implications.

The extracted analyst recalls validate our empirical approach and reproduce key principles of human memory. The analyst displays strong recency and temporal contiguity effects as she focuses more on recent episodes and tend to recall episodes that occurred close together in time. She also tends to recall salient episodes. For example, during the COVID-19 pandemic, the analyst recalled the 2008 financial crisis more than recent quarters. These patterns suggest that market participants weigh past experiences dynamically rather than following a fixed decay of attention over time. These results align with survey evidence ([Jiang et al., 2025](#)) and offer a better understanding of extrapolation and experience effect-based models of beliefs ([Barberis et al., 2015](#); [Malmendier and Nagel, 2011](#)).

Although forming beliefs based on recalls can be rational, deviations arise when the analyst retrieves the wrong historical episodes. We shed light on these deviations by constructing a real-time benchmark memory model and recalls, following [van Binsbergen et al. \(2023\)](#) and [Bianchi et al. \(2022\)](#). We apply the same two-step approach as for the analyst: first estimating the benchmark LSTM and its mental context, then deriving recalls based on similarity across benchmark contexts. The benchmark LSTM is trained on realized earnings revisions instead of consensus analyst forecast revisions. Its objective is to generate real-time optimal forecasts using the same information available at the time, rather than replicating analyst behavior. It approximates a “rational” model, with predictive power for realized earnings revisions comparable to non-memory baseline models and much higher than the analyst LSTM. Relative to this benchmark, analyst recalls exhibit significant distortions. In normal times, the analyst over-recalls distant episodes, relying too heavily on long-term

memory even when recent experiences are more relevant. During crises, by contrast, the analyst under-recalls distant episodes precisely when they are most informative.

Analyst recall distortions could arise from two mechanisms: feature encoding and selective forgetting, both grounded in psychology and economics. First, the analyst does not encode external information into memory efficiently. Comparing analyst and benchmark mental contexts, it shows that the analyst overweights past earnings and forecasts, while she should have put more emphasis on other firm fundamentals, as suggested by the benchmark. The analyst also adjusts feature importance insufficiently during crises, causing her to under-recall distant but informative episodes and underreact in the earnings forecast. Second, recall distortions also arise from selective forgetting, which refers to the selective retention of past information in memory due to limited memory capacity. The analyst applies this process differently from the benchmark. In a counterfactual analysis where we remove this mechanism from the LSTM, analyst recalls become more similar to the benchmark, displaying stronger recency and greater focus on positive episodes. In contrast, benchmark recalls show weaker recency and become more pessimistic.

Next, we show three sets of new asset pricing predictions based on analyst recalls and recall distortions.⁴ First, analyst recalls and their distortions predict both belief formation and future returns. When the analyst recalls positive (negative) episodes, defined as past episodes in which she would have revised forecasts upward (downward), she tends to form optimistic (pessimistic) beliefs. Stocks linked to positive recalls earn higher returns. Relative to benchmark recalls, overly positive analyst recalls are associated with over-optimistic forecasts (positive forecast errors) and short-term stock overpricing, while overly negative recalls predict the opposite. The mispricing reverses within about six months. These results suggest that the recalls extracted from our memory models contain useful information relevant for both the market participants' beliefs and the rational beliefs.

Second, we show that memory offers a powerful microfoundation for belief disagreement. Even when analysts share the same past experiences and current mental context, their recalled episodes can differ because memory retrieval is inherently stochastic (Kahana,

⁴All asset pricing tests are out-of-sample. For predicting asset pricing quantities at time t , our memory-based predictors are constructed using only information and models estimated before t .

2012; Bordalo et al., 2023). When these recalled episodes have heterogeneous outcomes (e.g., some positive, some negative), analysts form dispersed beliefs, generating disagreement. This mechanism further predicts higher abnormal trading volume in the associated stocks.

Third, our framework provides insights into asset pricing anomalies such as short-term reversal. When investors over-recall the prior month, they overweight last month’s return when forming beliefs about the current month, then buy past winners and sell losers, which attenuates short-term reversal. We measure this distortion using the difference between analyst and benchmark recall probabilities for the prior month. Short-term reversal turns into momentum among stocks with over-recall but is amplified among those with under-recall. The difference in short-term reversal performance between these two groups yields a risk-adjusted return of 1.09%, more than triple the standard short-term reversal effect.

The contribution of this paper to the literature is three-fold. First, this paper contributes to the literature linking human memory to economics and finance. Prior work has mainly developed theories connecting memory and financial markets (Bordalo et al., 2020; Nagel and Xu, 2022; Bordalo et al., 2023; Wachter and Kahana, 2024; Voigt, 2024). We provide new empirical evidence on the role of memory in belief formation by developing a structural approach to estimate recalls directly from market data. Our method yields firm-level analyst recalls spanning several decades, complementing survey and experimental studies as well as other empirical work focused on specific settings (Bordalo et al., 2025; Jiang et al., 2025; Enke et al., 2024; Gödker et al., 2025; Graeber et al., 2024; Charles, 2025). Our structural approach also allows us to explore distinct memory channels that shape recall and beliefs, such as feature encoding (Woodford, 2020; Frydman and Jin, 2022) and selective forgetting (Walters and Fernbach, 2021; Gödker et al., 2025). Finally, our results show patterns align with many other important reduced-form belief models (e.g., Malmendier and Nagel, 2011; Barberis et al., 2015; Bouchaud et al., 2019), suggesting that memory may provide a unifying cognitive foundation for them. In contemporaneous work, Charles and Sui (2024) propose a reduced-form approach that measures the similarity between episodes based on the objective similarity of external features. Our approach differs by deriving analyst recalls from subjective mental representations, using a psychologically grounded memory model. This allows us

to analyze the underlying mechanisms of belief formation and connect them to asset pricing phenomena.

Second, this paper provides novel insights into asset pricing patterns through the lens of memory-based belief formation, including return predictability (Barberis et al., 1998; Bordalo et al., 2019, 2024), disagreement and trading volume (Hong and Stein, 2007; Atmaz and Basak, 2018; Cookson and Niessner, 2020; Liao et al., 2022), and mispricing and asset pricing anomalies (Barberis, 2018; De la O et al., 2023; Da et al., 2014; Lehmann, 1990; Jegadeesh, 1990; Bouchaud et al., 2019; Cui et al., 2024).

Third, this paper adds to the growing literature applying machine learning and AI in finance and economics. Prior work uses these tools to forecast stock returns and economic outcomes such as EPS, GDP growth, and inflation (Gu et al., 2020; Chen et al., 2023; van Binsbergen et al., 2023; Bianchi et al., 2022; Lopez-Lira and Tang, 2023). We instead leverage machine learning models grounded in neuroscience and psychology to capture how market participants form beliefs and make decisions.⁵ Another strand of work uses embeddings which are vector representations of economic objects such as financial news, stock holdings, and firms, to capture stock returns and valuations (Chen et al., 2024; Gabaix et al., 2024; Sarkar, 2025). Recent work also uses large language models to analyze analyst reports and infer how they form beliefs (Bastianello et al., 2025; Ke, 2025). We draw on memory models in psychology, treating embeddings as analysts’ mental representations of firms’ states and using their similarity to measure recalls. This approach opens a new avenue for applying machine learning in behavioral finance. While similar ideas are emerging (e.g., Barberis and Jin, 2023), we are the first to implement them with empirical data.

The remainder of the paper is organized as follows. Section 2 presents our estimation procedure for analyst mental context and recalls. Section 3 reports the estimated analyst recalls and examines their relationship with analyst forecasts and market beliefs. Section 4 analyzes how analyst recalls are distorted relative to the benchmark and explores the

⁵Mullainathan and Spiess (2017) note challenges in interpreting machine learning models, related to the Rashomon effect that refers to the existence of many models that fit the data equally well but rely on different features, especially when inputs are highly correlated. We address this by focusing on higher-level patterns, such as the similarity of estimated vectors, rather than individual parameters, providing stable summaries across models.

resulting asset pricing implications. Section 5 investigates potential drivers of these recall distortions. Finally, Section 6 concludes.

2 Estimation of Mental Context and Recall

In this section, we present our estimation of mental context and recall in financial markets. We model the memory of a representative analyst, interpreted as the market’s collective memory for each firm. This framework allows us to study how memory shapes aggregate asset pricing patterns. Focusing on a representative analyst is both practical and conceptually grounded. Empirically, without rich individual-level data, estimating a representative memory process is most feasible (see Section 2.3). Conceptually, this focus parallels representative-agent models in asset pricing. Evidence from psychology literature also supports this aggregation: the key regularities of human memory, including similarity-based recall, aggregate well across individuals (e.g., Healey and Kahana, 2014).⁶ Extending the framework to incorporate richer heterogeneity in individual memory is a promising direction for future research. As a first step, Section 3.3.2 uses the representative analyst framework to link randomness in cued recall to disagreement and trading volume.⁷

The estimation of analyst mental context and recalls takes two steps. In the first step, we estimate the analyst mental context which is her perception of firms’ states. We use a structural memory model grounded in a neural network called Long-Short Term Memory (LSTM). In the second step, we extract the analyst recalls cued by the estimated mental context.

⁶For instance, Healey and Kahana (2014) document that recency, temporal contiguity, and semantic similarity effects are highly consistent across individuals. As they put it, these effects among individuals “could be characterized as a sample from a multivariate distribution described by the average function—that is, generated by the same underlying processes” (p. 590).

⁷Although analysts differ in personal experiences, past episodes often enter their memory through learning. For example, one analyst who graduated in 2016 assessed the potential impact of COVID-19 by writing: “As such, we believe it is useful to analyze what happened to live entertainment spend during the 1) 2008 global financial crisis; 2) H1N1 swine flu outbreak; and 3) the SARS coronavirus outbreak.” Such learning creates shared memory patterns among analysts, making a representative analyst a reasonable approximation of the market’s collective memory.

2.1 Step 1: Mental Context

We first introduce mental context in the dynamic memory structure, LSTM. Then we discuss why we use LSTM in our empirical setting and how we connect it with analyst forecasts.

2.1.1 The Structure of LSTM

LSTM is a type of neural network with a chain-like structure designed to handle sequential data. [Hochreiter and Schmidhuber \(1997\)](#) first introduced its theoretical foundation in machine learning literature. The structure of an LSTM unit naturally parallels the state-of-the-art memory model in psychology and economics by [Wachter and Kahana \(2024\)](#), as both are built around three core components. First, the external features $X \in \mathbb{R}^J$ capture public information such as macroeconomic conditions and firm fundamentals. Second, the memory cell $m \in \mathbb{R}^K$ functions as long-term memory, storing information over extended horizons and allowing earlier signals to influence later states. Third, the mental context $c \in \mathbb{R}^K$ serves as the current state representation, summarizing both new inputs and the information retrieved from memory. Here, K represents the dimension of the memory cell and mental context, and J denotes the number of external features. The flow of information is controlled by three gating mechanisms: a forget gate, an input gate, and an output gate. Based on the new external features together with the previous memory cell and mental context, the gates selectively determine which past information to retain in the memory, what new information to store, and what part of the updated memory to use for the current mental context.

Figure I illustrates the structure of one LSTM unit. The three gates, shown in orange boxes, are regulated by a sigmoid activation function σ , which applies an element-wise operation that outputs values between zero and one, controlling how much information passes through the gates. “tanh” in yellow boxes refers to the hyperbolic tangent activation function. These activation functions introduce non-linearity to better fit empirical economic and financial data and help further standardize variables to reduce the impact of outliers.⁸

⁸The operators “ $\times$ ” and “ $+$ ” shown in pink circles indicate element-wise multiplication and addition. The sigmoid function is defined as $\sigma(x) = \frac{1}{1+e^{-x}}$, and the hyperbolic tangent as $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$. Their inclusion also improves the accuracy and efficiency of model estimation ([Jagtap et al., 2020](#); [Dubey et al., 2022](#)).

The memory cell and the mental context are updated in the following steps. First, the forget gate determines the amount of information to retain in the memory cell and thereby controls selective forgetting. Specifically, an output of zero from the sigmoid function indicates that the information should be completely erased, whereas an output of one signals that it should be fully retained,

$$\text{forget}_t = \sigma(W_c^f c_{t-1} + W_x^f X_t + w_0^f). \quad (2.1)$$

Second, the input gate determines the extent to which new information is stored into the memory cell. If the sigmoid function outputs zero, the information is deemed irrelevant and is not stored; conversely, an output of one indicates that the information is fully stored in the memory cell. Before being stored, the information is transformed by the tanh function, which scales it to a range between -1 and 1, denoted as $\tilde{x}_t$:

$$\text{input}_t = \sigma(W_c^i c_{t-1} + W_x^i X_t + w_0^i) \quad (2.2)$$

$$\tilde{x}_t = \tanh(W_c^{\tilde{x}} c_{t-1} + W_x^{\tilde{x}} X_t + w_0^{\tilde{x}}). \quad (2.3)$$

Third, the memory cell is updated according to the outputs of the forget and input gates as follows:

$$m_t = \text{forget}_t \times m_{t-1} + \text{input}_t \times \tilde{x}_t, \quad (2.4)$$

by erasing contents from the memory cell and adding in new valuable information to it.

Finally, the output gate determines which information is relevant and should be elicited from the updated memory cell:

$$\text{out}_t = \sigma(W_c^o c_{t-1} + W_x^o X_t + w_0^o). \quad (2.5)$$

Then, the final output, the current mental context c_t , is chosen by the output gate from the updated memory cell through the tanh function:

$$c_t = \text{out}_t \times \tanh(m_t). \quad (2.6)$$

Note, W_x , W_c , and w_0 are model parameters that need to be estimated.

LSTM captures the core idea of the retrieved-context memory model in Wachter and Kahana (2024) that the mental context c_t should be dynamic and evolving endogenously,

going beyond just the static features of the physical environment. This is evident from two aspects of the model’s structure. The dynamics of the memory cell m_t and mental context c_t are analogous to those in the retrieved-context theory. In both frameworks, the current mental context is a function of the prior mental context and new information retrieved from a long-term memory store. This memory is then updated by accumulating new associations between the current mental context and observed external features. This creates a feedback loop where memory shapes the interpretation of new information, and that interpretation updates memory.

Second, the gates that govern information flow depend on both the prior mental context c_{t-1} and the new external features X_t , making memory updating context contingent. The model learns not only what information to retain and incorporate, but also how to encode it given the current state.⁹ For example, a reported 11% revenue growth (X_t) can be interpreted very differently based on the analyst’s perceived state of the firm (c_{t-1}). An analyst focused on a firm’s past difficulties may view the growth rate as evidence of recovery, while another who sees the firm in a high-growth state may regard the same number as a sign of slowing momentum.

LSTM has a recurrent architecture in which units are linked across time. Information stored in the memory cell and mental context at time $t - 1$ feed into processing at time t . The prior states (m_{t-1} and c_{t-1}) are used both to compute the gates that process new information and to update the memory cell and mental context. We illustrate this temporal recursion in Section 2.1.3.

Next, we explain why the LSTM provides a suitable framework for studying belief formation in financial markets.

⁹This mechanism generalizes the Kalman filter: in a Kalman filter, the transition and gain matrices are fixed (or evolve exogenously), whereas in LSTM the updating process are endogenously determined by the current mental context and external features. As a result, information integration is state-dependent rather than governed by a constant linear update rule.

2.1.2 Why LSTM?

The benefit of applying LSTM to study belief formation processes in financial markets is twofold. First, LSTM provides a psychologically grounded model of human memory, building on well-established frameworks in psychology and economics. Like the memory model of Wachter and Kahana (2024), LSTM captures three fundamental principles of human memory (Kahana, 2012): recency, temporal contiguity, and semantic similarity. Recency means that people refer to recently experienced episodes when accessing memory. Temporal contiguity means that people tend to recall an episode that occurred contiguously in time to presently-recalled episodes.¹⁰ Semantic similarity means that people are more likely to access the episodes that are most similar to that they are experiencing. Our framework thus allows us to illustrate their empirical patterns in financial markets.¹¹ The comprehensive structure of LSTM further lets us explore other memory mechanisms, such as feature encoding and selective forgetting (see Section 5).

Second, LSTM is well suited for processing high-dimensional and non-stationary financial variables with complex functional relationships, which analysts encounter in financial markets. Beyond modeling external features as basis vectors (Wachter and Kahana, 2024), LSTM as a machine learning approach offers advantages for handling these challenges, including embedded feature selection and dimension reduction (Nagel, 2021). It achieves this by using gates to selectively filter old information and weight new information. We apply LSTM on a broad set of external features, it then allows us to observe how the analyst assigns different weights to these features in memory over time (see Section 5.1).¹²

¹⁰For instance, Wachter and Kahana (2024) illustrate temporal contiguity in investor memory as follows. In their model, the 1930 Depression follows the 1929 stock market crash in memory. When investors experience a current financial crisis, their memory of 1929 and the subsequent Depression is cued, even if the features of crises and depressions are assumed unrelated. This shows how episodes close in time are recalled together due to temporal contiguity.

¹¹In our dynamic memory structure, the mental context c evolves according to the association of external stimuli, inner memory processes, and previous mental context. Howard and Kahana (2002) show that this autoregressive structure of c , embedded within the autoregressive memory cell m , naturally produces recency. The interaction of autoregressive mental context and the memory cell that stores long-term information establishes a channel for temporal contiguity. Semantic similarity emerges from contextually cued recall (see Section 2.2). Without a dynamic context structure, other memory frameworks that use static context embedded in the physical environment cannot fully capture these memory principles (Bordalo et al., 2020).

¹²LSTM has been shown to capture underlying dynamics in non-stationary variables. For example, Chen et al. (2023) demonstrate that LSTM can extract hidden states from cyclical macroeconomic variables, and

Conceptually, our memory framework can be viewed as an extension of Bayesian learning. Bayesian models update beliefs through fixed probabilistic rules given a prespecified set of states, features, and their associations. In high-dimensional empirical settings, it is often impossible to fully specify all relevant states and associations, making Bayesian updating difficult to implement. Memory models provide an adaptive alternative by forming and refining these associations endogenously, based on experiences of temporally linked episodes rather than on predefined structures.¹³

In machine learning and neuroscience, LSTM is recognized as a relatively simple but fundamental model. While more complex models exist, we apply LSTM in this paper to demonstrate that the memory mechanisms it captures show strong predictive and explanatory power for analyst beliefs and asset pricing patterns. Our approach can also be adapted to explore other specific memory or neuroscientific channels by replacing the basic LSTM model with more advanced models, such as those introduced by [Weston et al. \(2014\)](#), [Graves et al. \(2016\)](#), and attention-based models [Vaswani et al. \(2017\)](#).¹⁴

2.1.3 LSTM and Analyst Belief Formation

We now apply LSTM to the setting of analyst forecasts. Figure II presents the whole structure of how we model the analyst processes external features, updates memory and forms mental context using LSTM, then make forecasts for one firm as time progresses from left to right. For simplicity, the subscript i representing the firm is omitted in the figures, for example, $X_{i,t}$ is shown as X_t . In this structure, $X_{i,t} \in \mathbb{R}^J$ denotes the external features,

[Bianchi et al. \(2022\)](#) use LSTM to construct unbiased benchmarks for forecasts of GDP growth and inflation.

¹³See [Wachter and Kahana \(2024\)](#) for a detailed comparison between retrieved-context memory models and Bayesian learning frameworks.

¹⁴Our empirical approach of estimating analyst recalls is conceptually analogous to the attention mechanisms, as both weight past information by its relevance to the current context. In our two-step procedure, LSTM first generates a latent mental context summarizing the analyst’s current mental state, and recalls are then identified based on the similarity between this mental context and past mental contexts (see Section 2.2). In attention-based models, by contrast, relevance weights are computed directly from pairwise similarities without maintaining a distinct, dynamically evolving memory state. Our framework based on LSTM more closely resembles economic models of memory, with an explicit memory cell that evolves over time and allows us to study specific channels such as selective forgetting.

and $\Delta F_{i,t}^d$ denotes the direction of the forecast revision for firm i at time t :

$$\Delta F_{i,t}^d = \begin{cases} -1 & \text{if } F_{i,t}[EPS_{i,t+l}] < F_{i,t-1}[EPS_{i,t+l}], \\ 0 & \text{if } F_{i,t}[EPS_{i,t+l}] = F_{i,t-1}[EPS_{i,t+l}], \\ 1 & \text{if } F_{i,t}[EPS_{i,t+l}] > F_{i,t-1}[EPS_{i,t+l}], \end{cases} \quad (2.7)$$

where $F_{i,t}[EPS_{i,t+l}]$ is the consensus analyst forecast formed at time t for firm i 's future earnings per share $EPS_{i,t+l}$ that is available at time $t + l$, and $F_{i,t-1}[EPS_{i,t+l}]$ is the time $t - 1$ analyst consensus EPS forecast. $\Delta F_{i,t}^d = -1$, if the representative analyst revises down the forecast from last period; $\Delta F_{i,t}^d = 1$, if the analyst revises up the forecast; and $\Delta F_{i,t}^d = 0$, if the analyst does not change the forecast.¹⁵

The LSTM structure is recurrent: each unit passes its states forward over time. At date t , the inputs to the LSTM cell are the new information $X_{i,t}$ and the previous states $(m_{i,t-1}, c_{i,t-1})$. The cell updates these to produce the current memory cell $m_{i,t}$ and mental context $c_{i,t}$. The analyst then maps the mental context $c_{i,t}$ to the direction of the forecast revision $\Delta F_{i,t}^d$ via a multinomial logit function.

Both external features $X_{i,t}$ and the direction of forecast revisions $\Delta F_{i,t}^d$ are observable. We introduce the empirical data we use to construct $X_{i,t}$ and $\Delta F_{i,t}^d$ in Section 2.3. Thus, we can estimate the analyst memory cell $m_{i,t}$ and mental context $c_{i,t}$ by feeding external features $X_{i,t}$ into the whole structure and fitting revision directions $\Delta F_{i,t}^d$.

Additionally, we provide a benchmark memory model by training LSTM to fit the direction of realized EPS revisions which are analogously defined as shown in Equation (2.7):

$$\Delta RE_{i,t}^d = \begin{cases} -1 & \text{if } EPS_{i,t+l} < F_{i,t-1}[EPS_{i,t+l}], \\ 0 & \text{if } EPS_{i,t+l} = F_{i,t-1}[EPS_{i,t+l}], \\ 1 & \text{if } EPS_{i,t+l} > F_{i,t-1}[EPS_{i,t+l}]. \end{cases} \quad (2.8)$$

By comparing $F_{i,t-1}[EPS_{i,t+l}]$ with $EPS_{i,t+l}$, $\Delta RE_{i,t}^d$ is interpreted as the correct forecast revisions that the representative analyst should have made conditional on her forecast in the

¹⁵We focus on modeling the directions of forecast revisions because their small magnitudes reduce model training efficiency. This choice aligns with evidence that the magnitudes of analyst forecasts are noisy, while the direction provides a more robust measure for asset pricing patterns (Chiang et al., 2019). In later sections, we also include magnitudes of forecast revisions in our empirical tests.

preceding period. Although the model is trained on ex-post realized EPS revisions, we apply it strictly out of sample to avoid look-ahead bias. Its objective is to construct an empirical benchmark for optimal, real-time forecasts, using only the same information available to the analyst at the time. LSTM provides a suitable benchmark for this purpose, as it is capable of approximating optimal belief updating under complex, non-linear, and high-dimensional information environments by efficiently processing large sets of features. This is consistent with the machine-learning benchmark approach in [Bianchi et al. \(2022\)](#) and [van Binsbergen et al. \(2023\)](#). We later demonstrate that this benchmark memory model also serves as a reference for rational market beliefs. It not only helps capture biased analyst beliefs, but also predicts asset mispricing in financial markets.

2.2 Step 2: Cued Recall

Mental context is commonly used as a retrieval cue in the process of cued recall, which searches memory for past experiences similar to the current episode ([Wachter and Kahana, 2024](#); [Kahana, 2012](#)). We define an experience as a tuple (firm, month), and we refer to the retrieved past experiences as recalls. Retrieval is similarity based: past experiences that are closer to the current episode are more likely to be recalled. Given the estimated mental context c , we compute similarity between experiences and identify analyst recalls.

Specifically, for firm i at time t , the representative analyst has the memory database $M_{i,t}$, which consists of past experiences that are associated with the firms in the same industry as firm i before time t . Formally, $M_{i,t} \equiv \{(j, \tau)\}, \forall j, \tau$, such that $\text{Industry}(j) = \text{Industry}(i)$ and $\tau < t$. Then the similarity between any past experience in her memory database, e.g., $(j, \tau) \in M_{i,t}$, following standard methods in the memory literature ([Kahana, 2012](#)), is¹⁶

$$S(c_{j,\tau}, c_{i,t}) = \exp(-\|c_{j,\tau} - c_{i,t}\|_2). \quad (2.9)$$

¹⁶In general, the similarity function can be defined as

$$S(c_{j,\tau}, c_{i,t}) = \exp(-\xi \|c_{j,\tau} - c_{i,t}\|_\gamma),$$

where γ is the distance metric, with $\gamma = 2$ denotes the Euclidean norm, and $\xi \geq 0$ measures how quickly similarity decays with the distance. In our empirical tests, we take the most common case where $\gamma = 2$ and $\xi = 1$.

Two experiences are similar to each other if the Euclidean distance between their corresponding mental contexts is small. The similarity reaches a maximum of 1 when two mental contexts are exactly the same. The probability that a past experience (j, τ) is recalled, conditional on the current episode (i, t) , is proportional to their similarity:

$$p(c_{j,\tau}|c_{i,t}) = \frac{S(c_{j,\tau}, c_{i,t})}{\sum_{(m,s) \in M_{i,t}} S(c_{m,s}, c_{i,t})}. \quad (2.10)$$

These recalls define a similarity-based weighting over past episodes, determining how much each influences current beliefs. We then develop a recall-based framework of belief formation in Section 3, following [Bordalo et al. \(2025\)](#).

2.3 Data

Our data construction and processing follow [van Binsbergen et al. \(2023\)](#). We use a broad set of monthly public signals as external features X , including financial ratios from WRDS, firm fundamentals from COMPUSTAT, macroeconomic variables from the Federal Reserve Bank of Philadelphia, and earnings-related variables from I/B/E/S. Appendix Table A.1 lists all 79 features. The sample spans from January 1990 to December 2020, with approximately 1,300 firms per month on average. Industries are classified according to the Fama-French 49 portfolios. To limit the effect of outliers, all variables are winsorized at the 2.5 percent level cross-sectionally each month. To avoid look-ahead bias, all features in $X_{i,t}$ are publicly available before month t , mostly released in month $t - 1$.

We obtain data on EPS forecasts and realized EPS from I/B/E/S and process them following standard procedures in the literature ([Bouchaud et al., 2019](#); [Diether et al., 2002](#)). We get individual one-year-ahead EPS forecasts from the I/B/E/S Detail History File (unadjusted) and take the mean of these forecasts as the consensus. We then match realized EPS from the I/B/E/S unadjusted actuals file with the forecasts, using CRSP cumulative adjustment factors to put them on the same share basis.¹⁷

¹⁷The framework and empirical methods can also be applied to other cash flows, such as long-term earnings growth or dividends. While differences between short-term and long-term forecasts, and across cash flow types, are beyond the scope of this paper, understanding how memory affects these outcomes is an interesting avenue for future research.

Our analysis uses a comprehensive set of public signals that constitute the primary information set available to analysts. Although our framework could, in principle, incorporate private information, such data are not observable. We therefore rely on consensus forecasts to reduce the influence of unobservable private information and mitigate the impact of potential agency conflicts that may distort individual analysts’ forecasts. Although such conflicts may remain, they are unlikely to affect our main results for two reasons. First, prior work shows that such analyst incentives are unlikely to affect the time-series variation, which is the focus of our memory analysis (Bordalo et al., 2019). Second, our asset pricing results indicate that memory inferred from analyst forecasts captures beliefs relevant for marginal investors, consistent with prior literature (Bouchaud et al., 2019; van Binsbergen et al., 2023).

2.4 Estimation

We estimate two LSTMs using pooled data across all firms and the full market history: the analyst and benchmark LSTM models, trained on consensus analyst forecast revisions ΔF^d and realized revisions ΔRE^d , respectively. Practically, individual analyst forecasts are too noisy, sparse, and unevenly distributed across firms and time, making it infeasible to produce a well-trained model. Using consensus forecasts instead aggregates information and reduces idiosyncratic noise, allowing the model to learn stable and generalizable memory patterns.

We design the model training process, sample splitting, and performance evaluation following Gu et al. (2020). The base training sample spans January 1990 to December 2004, the validation sample January 2005 to December 2006, and the test sample January 2007 to December 2020. The validation sample is used to determine the hyperparameter K , the dimension of the memory cell and mental contexts, with the optimal value selected as $K = 10$. We employ a recursive expanding-window scheme to update the model and conduct out-of-sample analysis, which allows the model to adapt to changing environments and mimics how analysts update their recognition process over time. Further details are provided in [Appendix A.2](#).

Table I presents the out-of-sample performance of the analyst and benchmark LSTM models, covering both the validation and test samples from 2005 to 2020. For comparison, we also include a baseline multinomial logistic regression, which directly regresses $\Delta F_{i,t}^d$ and $\Delta RE_{i,t}^d$ on $X_{i,t}$, using the same recursive scheme and dataset as the LSTM models.

Table I compares the predictive accuracy of our LSTM models with a non-memory baseline in forecasting analyst revisions and realized earnings revisions. The analyst LSTM predicts analyst forecast revisions well and outperforms the non-memory multinomial logistic model. This highlights the importance of memory channels in shaping analyst belief formation. However, the analyst LSTM predicts realized earnings revisions poorly. This does not reflect a limitation of memory models: the benchmark LSTM approximates a “rational” model, with predictive power for realized earnings revisions similar to the baseline multinomial logistic. Taken together, these results suggest that using memory for belief formation is not inherently biased, but the analyst may apply it inefficiently.

To construct analyst LSTM recalls for firm i in month t , we proceed as follows. First, we obtain the most recently trained LSTM available prior to month t . We then input the external features $X_{i,t}$, which include public signals available before the start of month t , along with the previous mental context $c_{i,t-1}$ and memory cell $m_{i,t-1}$. The model produces the current mental context $c_{i,t}$, which serves as a retrieval cue to rank past experiences by similarity (2.9). In our empirical analysis, we consider a memory database $M_{i,t}$ of size five and define the five experiences that have the highest similarity to the cue as the analyst LSTM recalls for firm i in month t .¹⁸ The analyst LSTM recalls are therefore known before month t .

Analogously, we construct benchmark recalls using the benchmark LSTM trained prior to month t , applying the same external features $X_{i,t}$ as for analyst recalls. This ensures that the benchmark recalls are also generated in real time and out of sample, even though the benchmark LSTM is trained on realized earnings observed only ex post.

¹⁸Evidence from psychology shows that the number of items retrieved in cued and free recall tasks is limited (Earhard, 1967; Roberts, 1972), consistent with classic findings on working-memory capacity (Miller, 1956). Our results are robust to alternative memory database sizes, as shown in Appendix A.6.

3 Analyst Recall and Market Beliefs

In this section, we present the analyst LSTM recalls. We then develop a conceptual framework for recall-based belief formation that links recalls to forecasts, and show that these recalls have strong predictive power for analyst forecast revisions. Finally, we demonstrate that they also reflect market beliefs, as they predict stock returns and trading volume.

3.1 Analyst Recall

Figure III displays the analyst LSTM recalls. The darker blue areas indicate that, at the current time x (corresponding to the columns), experiences from time y (corresponding to the rows) are more likely to be recalled by the representative analyst. Specifically, at each time x , we identify the top five historical episodes most likely to be retrieved by the analyst for each firm in our sample, using the similarity function (2.9). Figure III plots the percentage of recalled episodes from time y across all firms analyzed at the current time x .

The findings are as follows. First, the analyst LSTM recalls validate our approach by reproducing well-established principles of human memory: the recency effect and the temporal contiguity effect. Specifically, the analyst tends to focus on recent episodes and to recall events that occurred close together in time. Because economic and financial variables are often persistent, such temporal structure could mechanically arise from autocorrelated external features rather than genuine memory linkages. To address this concern, we construct feature-based recalls (see Appendix A.4) directly from external features X , measuring similarity between the current episode (firm i , month t) and a past one (firm j , month τ) as

$$S = \exp(-\|X_{j,\tau} - X_{i,t}\|_2), \quad (3.1)$$

following Bordalo et al. (2020). While both feature-based and LSTM recalls display a strong recency effect, LSTM recalls also include more distant historical episodes and are more dispersed. Importantly, LSTM recalls uniquely exhibit richer temporal patterns such as contiguity and seasonality (episodes from the same quarter in the previous year are also frequently retrieved). To formally examine the temporal contiguity effect, we conduct a

simulation that removes autocorrelation in the external features (see [Appendix A.3](#)). The simulation confirms that analyst recalls exhibit strong temporal contiguity.¹⁹ Taken together, these findings suggest that the temporal linkages in analyst recalls reflect cognitive processes captured by our dynamic memory model rather than mechanical correlations among adjacent external features.

Second, consistent with the theoretical memory literature ([Wachter and Kahana, 2024](#)) and the experience effect literature ([Malmendier and Nagel, 2011](#)), we find that long-term memory and distant experiences are generally salient, sometimes even more influential. For example, during the COVID-19 pandemic in 2020, the analyst frequently recalled the 2008 global financial crisis (GFC), and its impact even dominated more recent episodes. Specifically, [Figure IV](#) shows the analyst recalls in 2020. In the second quarter of 2020, the analyst recalled the market crash period of the GFC. By the third quarter, rather than simply shifting her recall forward by one quarter, the analyst quickly shifted her attention to the GFC recovery period, which aligns with the timing of enforcement of the economic stimulus policies. By the fourth quarter of 2020, the analyst seldom recalled the GFC, considering the COVID pandemic unprecedented and concentrating instead on recent episodes. This overall recall pattern aligns with survey evidence in [Jiang et al. \(2025\)](#), which shows that investors are more likely to recall both recent and salient episodes.²⁰

Additionally, for each period, we calculate the frequency of the time difference between when a recall occurs and when the recalled experience took place. [Figure V](#) presents the average results for all periods from 2005 to 2020, along with specific averages for the 2008 GFC and the COVID-19 pandemic. Consistent with assumptions in the experience effect and extrapolation literature ([Malmendier and Nagel, 2011](#); [Nagel and Xu, 2022](#); [Barberis et al., 2015](#)), we find that, on average, the impact of past experiences decreases as the time difference grows. However, this assumption does not hold during certain periods. For instance, during the COVID-19 pandemic, the impact of the 2008 GFC remains prominent, as shown by

¹⁹The simulation results show that while the LSTM structure is capable of generating temporal contiguity, certain parameter settings can mute it. However, when trained on analyst forecasts, the model consistently displays the effect.

²⁰Surveys provide valuable direct evidence of people’s memory and recalls at specific points in time. Our approach complements this by showing, over time, which past episodes are recalled.

the 11th point on the orange line. Our estimated analyst recalls also address an essential question from the extrapolation literature: “How far back do people look when forming beliefs?” (Barberis, 2018). These findings, derived through our memory-based approach, suggest a more robust framework for modeling the impact of past experiences under different economic conditions.

3.2 Recall and Beliefs

3.2.1 A Conceptual Framework

To guide our empirical tests, we develop a conceptual framework for recall-based belief formation that links recalls to forecast revisions, following Bordalo et al. (2025) and Jiang et al. (2025).

At time t , the representative analyst (A) evaluates how to update her belief $F_{i,t}[EPS_{i,t+l}]$ about firm i ’s future EPS from her prior belief $F_{i,t-1}[EPS_{i,t+l}]$. Let $\Delta F_{i,t} = F_{i,t}[EPS_{i,t+l}] - F_{i,t-1}[EPS_{i,t+l}]$ denote the change in belief, that is, the forecast revision.

When forming $\Delta F_{i,t}$, the analyst may retrieve a past episode (j, τ) from memory. Each recalled episode serves as a reference, providing information about how forecasts were (or should have been) revised in similar situations. We define this information as the recalled revision $rr_{j,\tau}$:

$$rr_{j,\tau} = \begin{cases} EPS_{j,\tau+l} - F_{j,\tau-1}[EPS_{j,\tau+l}] & \text{if } \tau + l < t \\ F_{j,\tau}[EPS_{j,\tau+l}] - F_{j,\tau-1}[EPS_{j,\tau+l}] & \text{if } \tau + l \geq t \end{cases} \quad (3.2)$$

where $EPS_{j,\tau+l}$ denotes the realized earnings available at time $\tau + l$ and $F_{j,\tau-1}[EPS_{j,\tau+l}]$ is the corresponding forecast at time $\tau - 1$. When $rr_{j,\tau} > 0$, the analyst recalls a positive episode. This occurs either when she learns, based on realized earnings now available, that she should have revised her forecast upward in the recalled episode ($\tau + l < t$), or when she recalls having revised it upward at that time ($\tau + l \geq t$). When $rr_{j,\tau} < 0$, She recalls a negative episode.

Conceptually, this process parallels the “simulation” step in [Bordalo et al. \(2025\)](#), in which agents draw on retrieved episodes when imagining current outcomes. The current revision $\Delta F_{i,t}$ can be viewed as related to a similarity-weighted aggregation of recalled episodes:

$$\Delta F_{i,t} \propto \sum_{(j,\tau) \in M_{i,t}} p(c_{j,\tau}|c_{i,t}) \times rr_{j,\tau}, \quad (3.3)$$

where $p(c_{j,\tau}|c_{i,t})$ is the probability of recalling past episode (j, τ) , as defined in [\(2.10\)](#).²¹

3.2.2 Analyst Recall and Forecasts

Given the conceptual framework, we examine the validity of our framework of recall-based beliefs by testing the predictive power of analyst recalls for forecast revisions. Specifically, we examine Equation [\(3.3\)](#) using the following predictive regression

$$\Delta F_{i,t} = \beta \times RR_{i,t} + \varepsilon_{i,t}, \quad (3.4)$$

where $\Delta F_{i,t}$ is the consensus analyst EPS forecast revision scaled by the stock price $P_{i,t-1}$ at $t - 1$:

$$\Delta F_{i,t} = \frac{F_{i,t}[EPS_{i,t+l}] - F_{i,t-1}[EPS_{i,t+l}]}{P_{i,t-1}}. \quad (3.5)$$

$RR_{i,t}$ denotes the average recalled revision. We consider two specifications: $RR_{i,t}^A$ from analyst LSTM recalls and $RR_{i,t}^X$ from the feature-based recalls,

$$RR_{i,t}^A = \sum_{(j,\tau) \in M_{i,t}} p(c_{j,\tau}|c_{i,t}) \times rr_{j,\tau} / P_{j,\tau-1} \quad (3.6)$$

$$RR_{i,t}^X = \sum_{(j,\tau) \in M_{i,t}} p(X_{j,\tau}|X_{i,t}) \times rr_{j,\tau} / P_{j,\tau-1}, \quad (3.7)$$

where $p(X_{j,\tau}|X_{i,t})$ is the probability that a past experience is retrieved, proportional to the similarity function in Equation [\(3.1\)](#). Note that regression [\(3.4\)](#) is predictive because $RR_{i,t}$ is known at the end of month $t - 1$, as both $c_{i,t}$ and $X_{i,t}$ are available at that time.

Table [II](#) presents the regression results. Column (1) shows that analyst LSTM recalled revisions significantly predict forecast revisions, while Column (2) indicates that feature-

²¹Equation [\(3.3\)](#) also connects to the availability heuristic ([Tversky and Kahneman, 1974](#)) that people will be biased towards the instances that are easier to recall. In our setting, past episodes more similar to the current one receive higher weights when forming beliefs.

based recalled revisions lack similar predictive power. These results suggest that our model of analyst recalls based on LSTM more effectively captures key aspects of analyst belief formation than the feature-based recalls. In Column (4)-(6), we include three control variables: (i) lagged forecast revision $(F_{i,t-1}[EPS_{i,t+l}] - F_{i,t-2}[EPS_{i,t+l}])/P_{i,t-1}$; (ii) lagged earnings growth $(EPS_{i,t-l_1} - EPS_{i,t-l_2})/P_{i,t-1}$; and (iii) the lagged forecast $F_{i,t-1}[EPS_{i,t+l}]/P_{i,t-1}$, where $F_{i,t}[EPS_{i,t+l}]$ is the consensus analyst forecast for firm i 's future EPS at time t , $EPS_{i,t-l_1}$ and $EPS_{i,t-l_2}$ are realized earnings over the prior two fiscal years, and $P_{i,t-1}$ is the lagged stock price. All three controls are statistically significant, but their inclusion neither reduces the predictive power of LSTM recalls nor improves feature-based recalls. Overall, adding controls does not change our main findings.

3.3 Recall and Asset Pricing Implications

Our results show that the analyst recalls reflect her beliefs, and these recalls are constructed without using future stock prices or returns. The literature has shown that investors partially anchor their beliefs on analyst forecasts (Bouchaud et al., 2019; van Binsbergen et al., 2023). If our estimated analyst recalls capture this part of beliefs, we should expect they can predict asset pricing patterns in financial markets. We therefore test whether the analyst recalls are associated with stock returns and trading volume.

3.3.1 Recall and Stock Returns

First, we examine the link between analyst recalled revisions RR^A and stock returns by forming long-short portfolios. We hypothesize that positive (negative) recalled revisions predict future optimism (pessimism), leading to positive (negative) short-term returns. To reduce microstructure noise, we exclude stocks priced below \$5.

Panel A of Table III reports value-weighted, risk-adjusted returns of portfolios sorted into quintiles of RR^A , relative to the Carhart four-factor model (Carhart, 1997). The long-short portfolio generates a monthly return spread of 0.41% with a t -statistic above 3. Panel B shows that this return predictability reflects how investor beliefs align with analyst forecasts:

the analyst revises forecasts downward for firms in the lowest quintile and upward for firms in the highest quintile, and the difference is significant ($t = 7.27$). Appendix Table A.9 further shows that these long-short returns cannot be explained by standard asset pricing models. These findings provide evidence supporting our hypothesis that analyst recalls can predict market beliefs and thus exhibit return predictability.

3.3.2 Recall, Disagreement, and Trading Volume

Beyond return predictability, investor disagreement and trading volume are also central to asset pricing studies. However, less is known empirically about why there is disagreement among investors and thus trading volume. Literature on disagreement typically identifies two main sources: (i) differences in information sets, which include both current news and past experience, e.g., stored in each agent’s memory database, and (ii) different models applied to the same information set (Hong and Stein, 2007; Malmendier and Nagel, 2011; Cookson and Niessner, 2020). We provide a memory-based explanation for disagreement. The idea is that even with the same model and information set, different agents may retrieve distinct past experiences due to the inherent randomness in the process of cued recall (Bordalo et al., 2023).²²

Extending our framework from a representative analyst to individuals, the probability distribution $p(c_{j,\tau}|c_{i,t})$ in Equation (2.10) can be interpreted as the fraction of analysts or market participants who retrieve a given past experience (j, τ) . Then in this memory-based framework, disagreement arises under two conditions: (1) randomness in the process of cued recall, so that analysts may retrieve different past experiences from the same memory database, and (2) variation in the retrieved experiences, so that different analysts learn distinct signals.²³

²²Memory can also contribute to disagreement by creating unique information sets for each agent. Based on personal past experiences, agents form different memory databases, leading to distinct information sets or perceptions of the current episode (Bordalo et al., 2025).

²³To illustrate the idea, consider analysts evaluating firm i at time t , (i, t) . In the first case, if one past episode (j, τ) is overwhelmingly similar, everyone recalls it and forms the same belief, then there is no disagreement. In the second case, if two past episodes (j_1, τ_1) and (j_2, τ_2) are equally likely to be recalled but imply the same prospects, beliefs remain aligned. In the third case, if the two past episodes are equally likely but imply opposite signals (one positive, one negative), some analysts become optimistic while others pessimistic, generating disagreement.

If investor beliefs align closely with analyst beliefs, then disagreement among analysts should reflect disagreement among investors. Belief dispersion then leads to higher trading volume (Atmaz and Basak, 2018; Shalen, 1993). We therefore hypothesize that the dispersion of analyst recalls predicts both disagreement and trading activity. We test this using the following regressions:

$$\text{Disagreement}_{i,t} = \beta \times \sigma(rr)_{i,t} + \varepsilon_{i,t}, \quad (3.8)$$

and

$$\text{AbVol}_{i,t} = \beta \times \sigma(rr)_{i,t} + \varepsilon_{i,t}, \quad (3.9)$$

where recall dispersion $\sigma(rr)_{i,t}$ is the weighted standard deviation of the recalled revisions for firm i in month t ,²⁴

$$\sigma(rr)_{i,t} = \sqrt{\sum_{(j,\tau) \in M_{i,t}} p(c_{j,\tau}|c_{i,t}) \times (rr_{j,\tau}/P_{j,\tau-1} - RR_{i,t}^A)^2}. \quad (3.10)$$

This measure captures variation in recalled revisions (the second condition for disagreement) and weights each experience by its recall probability (the first condition).

Following Diether et al. (2002), we measure $\text{Disagreement}_{i,t}$ as the standard deviation of individual analyst forecasts for firm i in month t , scaled by the lagged stock price. AbVol is abnormal log trading volume, defined as log volume in month t minus the average log volume from $t - 12$ to $t - 2$, following Cookson and Niessner (2020). Regressions include firm and month fixed effects as well as controls for lagged abnormal volume, return, and volatility.

Table IV reports the results. Columns (1)–(3) show that recall dispersion significantly predicts forecast dispersion, though some observations are lost when fewer than two analysts issue forecasts. Columns (4)–(6) show a similar positive relation with trading volume. Overall, the evidence supports the hypothesis that dispersion in analyst recalls predicts both disagreement and trading activity, providing a memory-based microfoundation for disagreement.

²⁴Similar to $RR_{i,t}^A$, $\sigma(rr)_{i,t}$ is model-predicted, since it is based only on information before month t . The regressions (3.8) and (3.9) are therefore predictive.

4 Analyst Recall Distortions

Forming beliefs based on the recalled episodes can be rational. Deviation from full rationality arise when recalls are distorted, that is, when the analyst fails to retrieve the right historical episodes. In this section, we show that analyst recalls are indeed distorted relative to our benchmark recalls and discuss the associated asset pricing implications.

4.1 Recall Distortions

Figure VI presents the difference between the benchmark and analyst recalls. Historical periods that are more likely to be recalled by the benchmark are shaded in darker blue, while those are more likely to be recalled by the analyst are shaded in darker red. At each time x (corresponding to the columns), we identify the top five historical episodes most likely to be retrieved by the analyst and the benchmark for each firm. Figure VI then displays the difference in the percentage of episodes from time y (corresponding to the rows) that are recalled by the analyst versus the benchmark across all firms at current time x .

Compared with analyst recalls, benchmark recalls in each period are more concentrated. Generally, the benchmark focuses more on recent episodes, with distant past experiences largely deemed irrelevant, while the analyst is more influenced by long-term memory. Furthermore, analyst recalls sometimes are wrong. For instance, from 2010 to 2014, the analyst frequently recalled episodes from the boom period before 2008, while the benchmark recalls seldom referenced that period. Another example is from the COVID-19 pandemic, when the benchmark heavily focused on the 2008 GFC, analyst recalls were more dispersed.

The analyst also exhibits weaker recency effects in normal periods and stronger recency effects during crises compared with the benchmark. This suggests that the analyst does not fully adjust to environmental changes. In regular times, new input features prompt the benchmark to quickly update to recent, relevant episodes, while the analyst remains anchored in more distant experiences. During crises, the benchmark rapidly shifts focus, whereas the analyst underreacts, remaining influenced by recent episodes.²⁵

²⁵These patterns are consistent with De La O and Myers (2021), who show that analysts perceive the

We develop a formal statistical test to examine whether the analyst recalls differ from the benchmark.²⁶ We define recall distortion as

$$\Delta R_{i,t} = RR_{i,t}^A - RR_{i,t}^B, \quad (4.1)$$

where $RR_{i,t}^A$ and $RR_{i,t}^B$ denote the analyst and benchmark recalled revisions, respectively (Equation (3.6)). The null hypothesis that analyst recalls are not different from benchmark recalls corresponds to a mean recall distortion of zero, i.e., $\overline{\Delta R} = 0$. A paired t -test yields $t = -4.83$, indicating that analyst recalls are significantly distorted from benchmark recalls.²⁷

4.2 Recall Distortions and Asset Pricing Implications

We next examine the economic consequences of recall distortions, focusing on forecast errors, stock mispricing, and links to asset pricing anomalies.

4.2.1 Forecast Error

We first show that recall distortions capture analyst forecast errors. Analyst recalls strongly predict analyst forecasts (see Table II), if benchmark recalls capture the information relevant for realized earnings, then their difference should predict forecast errors. We test this implication using the following predictive regression:

$$e_{i,t} = \beta \times \Delta R_{i,t} + \varepsilon_{i,t}, \quad (4.2)$$

where the forecast error $e_{i,t} = (F_{i,t}[EPS_{i,t+l}] - EPS_{i,t+l})/P_{i,t-1}$ is the difference between the consensus forecast at time t and realized earnings, scaled by the stock price $P_{i,t-1}$. This regression is predictive because $\Delta R_{i,t}$ is computed using only information available before time t . We also include controls which are established predictors of forecast errors (Coibion

economic environment as more stationary than it actually is. We explore this mechanism further in Section 5.1.

²⁶The difference between analyst and benchmark recalls serves as a memory-based measure of inattention, consistent with defining inattention as deviation from a benchmark (Gabaix, 2019).

²⁷Standard errors are clustered at the industry and year levels.

and Gorodnichenko, 2015; Bouchaud et al., 2019; Bordalo et al., 2024): lagged forecast revision, lagged earnings growth, and lagged forecast, same as those in regression (3.4).²⁸

We also examine two alternative regression specifications. First, we focus on the magnitude, that is to test whether larger recall distortions lead to larger forecast errors, we take absolute values for both the regressor and the dependent variable:

$$|e_{i,t}| = \beta \times |\Delta R_{i,t}| + \varepsilon_{i,t}. \quad (4.3)$$

Second, instead of focusing on the magnitude, we analyze the direction of errors. We define forecast direction errors $e_{i,t}^d$ as an ordinal variable taking values $-2, -1, 0, 1, 2$:

$$e_{i,t}^d = \text{sgn}(\Delta F_{i,t}^d) - \text{sgn}(\Delta RE_{i,t}^d), \quad (4.4)$$

and estimate the effect of recall distortion on $e_{i,t}^d$ using an ordered logit regression.²⁹

Table V reports the results. Across all specifications, recall distortion significantly predicts forecast errors. Columns (3)–(4) show that larger distortions generate larger errors, while Columns (5)–(6) show that overly positive (negative) recalls lead to overly optimistic (pessimistic) beliefs. Columns (1)–(2), which combine these specifications, confirm that recall distortion robustly predicts forecast errors. Moreover, Columns (2), (4), and (6) demonstrate that even when competing with other forecast error predictors, recall distortion remains significant, and the magnitude of the coefficients remains relatively unchanged.

²⁸For example, Bordalo et al. (2019) run the regression:

$$EPS_{i,t+l} - F_{i,t}[EPS_{i,t+l}] = \beta_0 + \beta_1(F_{i,t}[EPS_{i,t+l}] - F_{i,t-1}[EPS_{i,t+l}]) + \beta_2 F_{i,t-1}[EPS_{i,t+l}] + \varepsilon_{i,t},$$

thus one of their predictors, forecast revision $F_{i,t}[EPS_{i,t+l}] - F_{i,t-1}[EPS_{i,t+l}]$ is known only at time t . Since we aim to predict forecast errors using information available before t , we include the lagged forecast revision instead of the current revision.

²⁹The ordered logit regression is

$$\Pr(e_{i,t}^d = j) = \Lambda(\kappa_j - x_{i,t}) - \Lambda(\kappa_{j-1} - x_{i,t}), \quad j \in \{-2, -1, 0, 1, 2\}, \quad (4.5)$$

where $x_{i,t} = \beta \times \Delta R_{i,t} + \theta \times Z_{i,t} + \gamma_i + \eta_t$ and $\Lambda(z) = 1/(1 + \exp(-z))$. The cutoffs κ_j are estimated jointly with the coefficients.

4.2.2 Subsequent Stock Returns

Taking this one step further, if analyst recalls capture market beliefs and benchmark recalls approximate rational beliefs, then recall distortions should reflect not only analyst forecast errors but also temporary asset mispricing. In an efficient market, such mispricing should eventually be corrected. Therefore, we expect a reversal in portfolio returns sorted on analyst recall distortions.

To test this, we sort stocks into quintile portfolios based on recall distortion ΔR . The first quintile contains stocks with overly positive analyst recalls (average $\Delta R = 2.80$), while the fifth quintile contains stocks with overly negative analyst recalls (average $\Delta R = -4.74$). We construct a value-weighted long-short portfolio that goes long in the fifth quintile and short in the first.

Figure VII plots the cumulative returns of this portfolio with 95% confidence intervals in the shaded region. Table VI reports the corresponding gaps in analyst forecast revisions between the two groups. Cumulative returns are negative in the beginning and reach a maximum of -0.51% around two months after portfolio formation, as the group with overly negative analyst recalls downwards forecasts more than the overly positive group during this period. Then, cumulative returns reverse thereafter and stabilize around zero after six months, consistent with the closing gap in forecast revisions across the two groups. Taken together, these results indicate that benchmark recalls approximate rational beliefs in the market and can help capture temporary asset mispricing.

4.2.3 Recall of the Prior Month and Short-Term Reversal

So far we have focused on one property of recalls, which is their positivity or negativity, measured by the forecast revisions the analyst should have made in the recalled episodes. However, recalls have many other characteristics that can be explored to offer new asset pricing insights. In this section, we show one example and examine their temporal aspects, which reveal the specific historical moments that the analyst is paying attention to at a given time. Comparing these temporal patterns with the benchmark recalls allows us to identify

whether the analyst over-recalls or under-recalls certain periods. This perspective provides a memory-based channel for understanding anomalies linked to temporal features, such as the short-term reversal. More broadly, our empirical approach highlights the potential to connect other recall characteristics to other anomalies.³⁰

Short-term reversal describes the phenomenon that a stock’s return over the last month is negatively associated with its subsequent return (Lehmann, 1990; Jegadeesh, 1990). We link this pattern to the temporal structure of recall. When investors over-recall the prior month, they perceive the current period as highly similar to the previous one. As a result, their beliefs about this month’s return load heavily on last month’s performance. Following a positive prior-month return, such investors expect continued gains and buy the stock, pushing prices up and attenuating the short-term reversal effect. Conversely, when investors under-recall the prior month, they view the current period as less related to the past and place less weight on the prior return, strengthening the reversal. As these recall distortions are measured relative to the benchmark, the efficient market should subsequently correct them, implying a reversal of the pattern in later months.

We measure biased investor recalls toward the prior month as the difference between analyst and benchmark recall probabilities:

$$\Delta\rho_{i,t} = \rho_{i,t}^A - \rho_{i,t}^B, \quad (4.6)$$

where $\rho_{i,t}^A$ is the likelihood that the analyst recalls episodes from month $t - 1$:

$$\rho_{i,t}^A = \sum_{(j,\tau) \in M_{i,t}} p(c_{j,\tau} | c_{i,t}) \times \mathbb{1}_{\tau=t-1}, \quad (4.7)$$

and $\rho_{i,t}^B$ is defined analogously for the benchmark. A positive (negative) $\Delta\rho_{i,t}$ indicates that the analyst over-recalls (under-recalls) the prior month relative to the benchmark, capturing the extent to which investors collectively over-recall (under-recall).

We hypothesize that over-recall weakens short-term reversal, while under-recall amplifies it. In an efficient market, we also expect these patterns to reverse in subsequent months.

³⁰For example, Kahana et al. (2025) provides a memory-based explanation for the value premium. Using our approach, one could also study other anomalies linked to temporal features, such as momentum and long-term reversal. We do not explore them here because these anomalies are relatively muted in our sample period (2007–2020).

To test our hypotheses, we examine the double-sorted portfolios. We first sort stocks into terciles by $\Delta\rho$. Median values of $\Delta\rho$ for each tercile are 0.18, -0.11, and -0.30, corresponding to over-recall in the first group and under-recall in the second and third. Within each $\Delta\rho$ tercile, we further sort firms by prior-month returns.

Table VII reports Carhart four-factor alphas for the double-sorted portfolios. For stocks exhibiting the strongest over-recall, the short-term reversal alpha is -0.58% per month, turning reversal into momentum: stocks with higher returns last month continue to earn higher returns this month as investors over-recall the previous month and overweight its implication for current beliefs. By contrast, for stocks with under-recall, the alpha is 0.51% , larger than the standard short-term reversal alpha of 0.34% . In this group, the reversal effect is amplified as investors believe this month is different from last month.³¹

The difference in short-term reversal performance between the $\Delta\rho$ groups is highly significant. The double-sorted long-short portfolio earns alpha of 1.09% ($t = 3.88$), more than triple the standard short-term reversal effect. Figure VIII further shows that this distortion is corrected within five months, as the cumulative returns of the double-sorted long-short portfolio revert to zero. These patterns indicate that memory offers new insights into the short-term reversal anomaly.³²

5 Driving Factors of Analyst Recall Distortions

So far, we have examined the economic consequences of analyst recall distortions through several asset pricing tests. But what drives these distortions? To explore this question, we consider two potential driving forces: feature encoding and selective forgetting.

³¹We re-estimate the standard short-term reversal strategy following Jegadeesh (1990) in Appendix Table A.10. In our sample, the Carhart-adjusted alpha is 0.34% per month, close to Da et al. (2014).

³²These findings extend prior work on extrapolation (Barberis et al., 2015; Barberis, 2018) by identifying over- and under-extrapolation of the previous month’s return. Our results are consistent with temporary mispricing driven by investor extrapolation, as over-extrapolated stocks subsequently exhibit momentum. Prior studies interpret extrapolation as placing excessive weight on recent performance and propose time-varying measures (Cassella and Gulen, 2018), whereas our benchmark-based framework quantifies this degree in real time and across stocks.

5.1 Feature Encoding

We investigate the first potential driving force by comparing how the analyst’s memory cell and mental context emphasize external features relative to the benchmark. Since both elements are central to memory models, examining how they manifest in empirical data sheds light on underlying memory mechanisms and helps guide future memory modeling.

5.1.1 Decomposition of the Memory Cell and Mental Context

Because the mental context and memory cell are latent and not directly interpretable, we study their relation to external features by summarizing the variable importance for each. Specifically, we examine how the importance of each external feature j at time t , denoted $\omega_{j,t}$, evolves over time, following [Gu et al. \(2020\)](#). To construct $\omega_{j,t}$, we proceed in three steps. First, for each of the K dimensions of the memory cell (m) and mental context (c), we run a linear regression on all external features using a 12-month rolling window. Second, we set feature j to zero while holding other features and coefficients fixed, and compute the reduction in R^2 as $\omega_{j,t}$. In the third step, to illustrate variable importance more clearly, we aggregate the 79 external features into four groups and report the sum of $\omega_{j,t}$ within each group, which we call group variable importance.³³ The four groups correspond to the panels in [Appendix Table A.1](#): (i) ω_t^C , firm characteristics other than earnings-related variable; (ii) ω_t^M , macroeconomics conditions; (iii) ω_t^R , market conditions; (iv) ω_t^F , historical earnings and forecasts.

Figures [IXa - IXd](#) present time-series plots of the group variable importance for the benchmark mental context and memory cells, as well as those for the analyst. The findings are twofold. First, variable importance changes over time for both the benchmark and the analyst.³⁴ For example, during recessions, macroeconomic variables become more important, while firm characteristics and historical earnings and forecasts become less relevant. This

³³Focusing on group variable importance, rather than individual variables, helps mitigate instability from the Rashomon effect ([Mullainathan and Spiess, 2017](#)). By aggregating correlated inputs, our results are less sensitive to which specific feature the model emphasizes.

³⁴Time-varying variable importance also aligns with textual evidence from analyst reports ([Bastianello et al., 2025](#); [Ke, 2025](#)).

pattern aligns with the intuition that recessions have market-wide effects and is consistent with limited attention theory (Kacperczyk et al., 2016) which suggests investors allocate more attention to aggregate news during downturns. Importantly, this shift does not imply bias, as the benchmark shows a similar response. This evidence supports the use of a dynamic memory model and highlights that capturing time-varying feature weighting is essential for understanding belief formation in financial markets.

Second, the analyst and the benchmark focus on different aspects. The benchmark places the most weight on firm characteristics in both memory cells and mental context, whereas the analyst overweights historical earnings-related variables. This distortion in how external features are encoded provides empirical evidence and a memory-based microfoundation for models of encoding errors (“early noise”) that links noisy internal representations of data to economic outcomes (e.g., Woodford, 2020; Frydman and Nunnari, 2021; Frydman and Jin, 2022; Drugowitsch et al., 2016). The evidence that the analyst overweights her own past forecasts reconciles with sticky expectations (Coibion and Gorodnichenko, 2015; Bouchaud et al., 2019), self-herding (Hirshleifer et al., 2019), and confirmation bias (Lord et al., 1979; Barberis, 2003). The reliance on past earnings growth is also consistent with fundamental extrapolation (Barberis et al., 1998, 2015). More broadly, our results suggest that memory potentially can be a unifying framework for key assumptions about beliefs in financial markets.

By comparing the decomposition of analyst and benchmark mental contexts, we assess the analyst misreaction in perceptions, that is, how the analyst responds to changes in the economic environment relative to the benchmark. We measure this using a double-differential approach.

First, we define the time-series change in group variable importance:

$$\Delta\omega_t = \sqrt{\frac{(\omega_t^F - \omega_{t-1}^F)^2 + (\omega_t^M - \omega_{t-1}^M)^2 + (\omega_t^R - \omega_{t-1}^R)^2 + (\omega_t^C - \omega_{t-1}^C)^2}{4}}, \quad (5.1)$$

where ω_t^F , ω_t^M , ω_t^R , ω_t^C represent the variable importance of the four feature groups. We compute $\Delta\omega_t$ separately for the benchmark ($\Delta\omega_t^b$) and the analyst ($\Delta\omega_t^a$).

Second, we measure the difference between the analyst and the benchmark:

$$\Delta^2\omega_t = \Delta\omega_t^b - \Delta\omega_t^a, \quad (5.2)$$

which captures how the analyst's perception of the change in the economic environment deviates from the benchmark. Positive values indicate underreaction, while negative values indicate overreaction.

Figure X shows that during normal periods, the analyst perceptions closely track the benchmark. During crises, their perceptions diverge. For example, during COVID-19, the analyst largely underreacted, failing to fully recognize the crisis's severity and the impact of subsequent economic stimulus.

5.1.2 Impact of Misreaction in Perceptions

We next examine how feature encoding biases the analyst recalls and forecasts by linking misreaction in perceptions to forecast errors and the recency effect in recalls.

First, we measure misreaction in forecasts following Coibion and Gorodnichenko (2015).

Let

$$\Delta\text{React}_{i,t} = \begin{cases} \text{sgn}(\Delta F_{i,t} - \Delta RE_{i,t}) & \text{if } \Delta F_{i,t} > 0 \\ \text{sgn}(\Delta RE_{i,t} - \Delta F_{i,t}) & \text{if } \Delta F_{i,t} < 0 \\ -1 & \text{if } \Delta F_{i,t} = 0 \end{cases}$$

where $\Delta F_{i,t} = (F_{i,t}[EPS_{i,t+l}] - F_{i,t-1}[EPS_{i,t+l}])/P_{i,t-1}$ denotes the analyst forecast revision and $\Delta RE_{i,t} = (EPS_{i,t+l} - F_{i,t-1}[EPS_{i,t+l}])/P_{i,t-1}$ denotes the realized EPS revision. Negative (positive) values of $\Delta\text{React}_{i,t}$ indicate underreaction (overreaction) in forecasts. To align with the monthly frequency of misreaction in perceptions $\Delta^2\omega_t$, we average $\Delta\text{React}_{i,t}$ across firms:

$$\Delta\text{React}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} \Delta\text{React}_{i,t}, \quad (5.3)$$

where N_t is the number of firms in month t .

Second, we measure biased recency in analyst recalls. It captures how likely recalls fall

within the last 12 months:

$$\text{Recency}_{i,t}^A = \sum_{(j,\tau) \in M_{i,t}} p(c_{j,\tau} | c_{i,t}) \times \mathbb{1}_{t-\tau \leq 12}. \quad (5.4)$$

The benchmark recency effect $\text{Recency}_{i,t}^B$ is defined similarly. The biased recency effect is then the monthly cross-sectional average of the difference:

$$\Delta \text{Recency}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} (\text{Recency}_{i,t}^B - \text{Recency}_{i,t}^A) \quad (5.5)$$

Table VIII presents regressions linking misreaction in perceptions $\Delta^2 \omega_t$ to misreaction in forecasts and biased recency effect. Column (1) shows that when the analyst underreacts to changes in the economic environment ($\Delta^2 \omega_t > 0$), her forecasts also underreact relative to realized EPS. When the analyst is slow to adjust the weights of external variables in the encoding process, she develops distorted perceptions, leading to misreactions in her forecasts. Columns (2) and (3) show that when the analyst underreacts to changes in the economic environment, she underweights recent episodes during normal periods, but overemphasizes them during crises. These results indicate that the patterns in Figure VI in which recent periods are underweighted in normal times and overemphasized during crises, can stem from the analyst’s insufficient adjustment to changes in the economic environment.³⁵

5.2 Selective Forgetting

For the second potential driving force, we examine how selective forgetting shapes recalls and beliefs. Selective forgetting is a well-studied memory process in psychology (Anderson and Hulbert, 2021), which refers to the selective retention of past information in memory due to limited memory capacity. Prior evidence in finance shows that it can bias investor decision-making, as individuals tend to recall positive outcomes more readily than negative ones (Walters and Fernbach, 2021; G dker et al., 2025).³⁶ We extend this literature by providing

³⁵We provide robustness checks in Appendix A.6. For ΔReact_t , we exclude cases where $\text{sgn}(\Delta F_{i,t}) \neq \text{sgn}(\Delta RE_{i,t})$, as they indicate the analyst revises forecasts in the wrong direction. These cases are included in the robustness check. For $\Delta \text{Recency}_t$, we also test alternative windows of three or five years. The main patterns remain unchanged.

³⁶Walters and Fernbach (2021) find that participants are more likely to forget losing trades than winning ones. G dker et al. (2025) document a similar asymmetry, with individuals over-remembering positive outcomes and under-remembering negative ones.

comprehensive empirical evidence on selective forgetting in analyst and benchmark belief formation.

To assess the role of selective forgetting, we conduct a counterfactual analysis that blocks this channel in the original model. We then examine how analyst recalls change and whether forecasts improve. In the full LSTM (2.4), the forget gate determines what content to retain in the memory cell. We remove this gate by fixing $\text{forget}_t \equiv \mathbf{1}$, which imposes full retention from previous memory cell, as shown in Figure XI. The modified cell then updates only by adding new information:

$$m_t = m_{t-1} + \text{input}_t \times \tilde{x}_t, \quad (5.6)$$

In this setup, the analyst cannot selectively forget. But this does not mean perfect recall. The finite dimension of memory cells implies that they remain subject to limited memory capacity and passive forgetting, and the new information gradually overrides older content. For instance, without selective forgetting, a large transitory shock can dominate the memory cell and make older signals effectively disappear. With selective forgetting, the model can instead down-weight this transitory shock, making it easier to recover underlying fundamentals.

We then take the originally trained full LSTM for both the analyst and the benchmark, replace the memory update process (2.4) with the no-forget version (5.6), and keep all other components and parameters unchanged. This modification yields new mental context and recalls, which we define as counterfactual recalls.

Figure XII compares the analyst LSTM recalls (shown in Section 3.1) to the counterfactual recalls. Relative to the counterfactual, darker blue areas indicate episodes that the analyst under-recalls, while the darker red areas show episodes that analyst over-recalls. Combined with the formal tests in Table IX, this comparison yields three main findings.

First, the analyst recalls without selective forgetting show a stronger recency effect. In Panel A of Table IX, Recency (5.4) rises from 0.052 to 0.104, and the average recalled revision, RR (3.6), increases from -0.707 to -0.625. The 0.082 difference is statistically significant ($t = 6.69$), indicating that the analyst recalls are more positive without selective forgetting. Figure XII also shows that these counterfactual recalls place greater weight on recent episodes, especially those within the past 12 months.

Second, without selective forgetting, analyst recalls are closer to the benchmark. The average analyst recalled revision increases to -0.625 without selective forgetting, closer to the average benchmark recalled revision of -0.434. Similarly, the recency effect increases to 0.104, approaching the benchmark recency of 0.177.

Third, selective forgetting affects the analyst and benchmark differently. Panel B of Table IX shows that removing selective forgetting makes benchmark recalls more pessimistic and reduces their recency effect, opposite to its effect on the analyst recalls.³⁷

In summary, selective forgetting is an important part of belief formation. However, the analyst applies this process differently from the benchmark, leading to recall distortions.

6 Conclusion

The psychology and finance literature highlights the role of memory in belief formation, yet field evidence from financial market data remains limited. This paper extracts two fundamental components of memory, mental context and recall, for a large cross-section of firms over several decades. We do so by empirically estimating a structural, memory-based model of a representative analyst forming earnings forecast revisions. Using this framework, we provide novel evidence linking memory to belief formation in financial markets. Analyst recalls deviate systematically from those implied by a benchmark memory model: the analyst relies too heavily on long-term memory and underreacts to new information, over-recalling distant episodes in normal times and under-recalling them in crises. We find that well-established memory principles, such as recency and temporal contiguity, and other memory channels including feature encoding and selective forgetting shape analyst recalls and beliefs.

Building on these findings, we derive new asset pricing predictions grounded in memory. Analyst recalls and distortions predict stock returns, trading volume, forecast errors, mispricing, and anomalies such as short-term reversals.

³⁷Table A.3 shows that without selective forgetting, the benchmark model’s predictive power for realized EPS revisions declines, with within R^2 falling by 40%, while for the analyst, predictive power rises, with within R^2 increasing by 68%. Thus, removing selective forgetting improves the analyst forecasts but worsens the benchmark’s performance. The overall within R^2 levels are relatively small, so this evidence should be interpreted as suggestive.

This framework opens new directions for research on memory and belief formation in finance and economics. First, with rich individual-level data, it can be readily extended to study heterogeneity in agents' memory and its asset pricing implications. Second, combining this structural approach with text-based evidence from analyst reports using large language models offers a complementary way to study belief formation. Third, it provides a foundation for studying memory as a unifying mechanism across belief-formation models. Fourth, its ability to handle complex and high-dimensional data makes it suitable for studying memory across a range of domains, including macroeconomic forecasts, consumer expectations, and managerial decisions. Finally, by drawing on the neuroscience and psychology foundations of certain machine learning models, our framework can be extended to explore how cognitive mechanisms beyond memory shape the belief formation process.

Tables and Figures

Table I: Out-of-Sample Prediction Accuracy of LSTM and Logistic Regression

Model	Analyst Forecast Revision (ΔF^d)	Realized Earnings Revision (ΔRE^d)
LSTM (Analyst)	56.68%	35.31%
LSTM (Benchmark)	37.37%	59.52%
Logistic	48.21%	58.46%

This table reports the average out-of-sample prediction accuracy for the trained LSTM model and the multinomial logistic regression. Accuracy is measured for two dependent variables: analyst forecast revisions (ΔF^d ; see (2.7)) and realized earnings revisions (ΔRE^d ; see (2.8)). Out-of-sample performance is evaluated using a recursive (expanding-window) scheme, as detailed in [Appendix A.2](#). The sample period is January 2005 to December 2020.

Table II: Analyst Forecast Revisions and Recalled Revisions

	Forecast Revisions ($\Delta F_{i,t}$)					
	(1)	(2)	(3)	(4)	(5)	(6)
LSTM recalled revision RR^A	0.059*** (0.015)		0.060*** (0.013)	0.061*** (0.013)		0.061*** (0.011)
Feature-based recalled revision RR^X		0.006 (0.013)	-0.005 (0.012)		0.010 (0.012)	-0.001 (0.011)
Lagged forecast revision				0.127** (0.043)	0.139*** (0.043)	0.127** (0.043)
Lagged earnings growth				0.013** (0.005)	0.014** (0.005)	0.013** (0.005)
Lagged forecast				-4.682*** (0.782)	-4.557*** (0.709)	-4.679*** (0.756)
Firm fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	277,410	277,410	277,410	277,410	277,410	277,410
R-squared	0.053	0.047	0.053	0.070	0.064	0.070
Within R-squared	0.006	0.000	0.006	0.025	0.018	0.025

This table reports estimates from regressions of the form

$$\Delta F_{i,t} = \beta \times RR_{i,t} + \theta \times Z_{i,t} + \gamma_i + \eta_t + \varepsilon_{i,t},$$

where $\Delta F_{i,t}$ is the consensus analyst forecast revision (defined in (3.5)) and $RR_{i,t}$ is the recalled revision. We consider two measures of $RR_{i,t}$: $RR_{i,t}^A$, the recalled revision based on the analyst LSTM model (defined in (3.6)), and $RR_{i,t}^X$, the recalled revision based on external features (defined in (3.7)). $Z_{i,t}$ is a vector of control variables, and γ_i and η_t denote firm and time fixed effects, respectively. Columns (1) to (3) are specifications without controls. Columns (4)-(6) add three controls: (i) lagged forecast revision ($F_{i,t-1}[EPS_{i,t+l}] - F_{i,t-2}[EPS_{i,t+l}]/P_{i,t-1}$); (ii) lagged earnings growth ($EPS_{i,t-l_1} - EPS_{i,t-l_2}/P_{i,t-1}$); and (iii) the lagged forecast $F_{i,t-1}[EPS_{i,t+l}]/P_{i,t-1}$, where $F_{i,t}[EPS_{i,t+l}]$ is the consensus analyst forecast for firm i 's future EPS at time t , $EPS_{i,t-l_1}$ and $EPS_{i,t-l_2}$ are realized earnings over the prior two fiscal years, and $P_{i,t-1}$ is the lagged stock price. The sample period is January 2007 to December 2020. Standard errors are two-way clustered by industry and year and are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table III: Recalled Revisions, Forecast Revisions, and Stock Returns

	1 (Low)	2	3	4	5 (High)	5 – 1
<i>Panel A: Risk-adjusted Returns</i>						
Mean	-0.06	0.17	0.13	0.26	0.35	0.41
	(-0.64)	(1.68)	(1.46)	(2.84)	(3.11)	(3.01)
<i>Panel B: Forecast Revisions</i>						
Mean	-0.14	-0.04	-0.01	0.04	0.17	0.31
	(-3.04)	(-1.52)	(-0.29)	(3.43)	(5.14)	(7.27)

This table presents average risk-adjusted returns (Panel A) and average analyst forecast revisions (ΔF , Panel B) for value-weighted portfolios. Portfolios are sorted into quintiles based on the analyst recalled revision (RR^A , defined in (3.6)). Risk-adjusted returns are reported in percent and are estimated using the Carhart four-factor model (Carhart, 1997). The sample period is January 2007 to December 2020. The t -statistics reported in parentheses are robust to heteroskedasticity and autocorrelation with up to 12 lags.

Table IV: Recall dispersion, Disagreement, and Trading volume

	Disagreement _{<i>i,t</i>}			Abnormal volume AbVol _{<i>i,t</i>}		
	(1)	(2)	(3)	(4)	(5)	(6)
$\sigma(rr)_{i,t}$	0.019*** (0.004)	0.017*** (0.003)	0.008*** (0.002)	0.089*** (0.028)	0.089*** (0.027)	0.038** (0.014)
$\sigma(rr)_{i,t-1}$		0.014*** (0.003)	0.006** (0.002)		0.001 (0.017)	-0.004 (0.019)
Disagreement _{<i>i,t-1</i>}			0.540*** (0.017)			
AbVol _{<i>i,t-1</i>}						0.539*** (0.010)
$Ret_{i,t-1}$	-0.009*** (0.002)	-0.010*** (0.002)	-0.012*** (0.002)	-0.044 (0.052)	-0.044 (0.052)	-0.086** (0.035)
$Ret_{i,t-2}$	-0.008*** (0.001)	-0.008*** (0.001)	-0.004*** (0.001)	-0.028 (0.039)	-0.028 (0.039)	0.004 (0.019)
$Ret_{i,t-3}$	-0.008*** (0.001)	-0.007*** (0.001)	-0.004*** (0.001)	0.033 (0.044)	0.033 (0.044)	0.055** (0.025)
$\sigma(Ret)_{i,t-1}$	0.078*** (0.012)	0.077*** (0.012)	0.039*** (0.006)	-0.133 (0.160)	-0.133 (0.160)	-0.573*** (0.110)
Firm fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	101,438	101,438	59,160	194,444	194,444	194,444
R-squared	0.435	0.437	0.609	0.175	0.175	0.412

This table reports regressions of the form

$$\text{Disagreement}_{i,t} = \beta \times \sigma(rr)_{i,t} + \theta \times Z_{i,t} + \gamma_i + \eta_t + \varepsilon_{i,t}$$

and

$$\text{AbVol}_{i,t} = \beta \times \sigma(rr)_{i,t} + \theta \times Z_{i,t} + \gamma_i + \eta_t + \varepsilon_{i,t}.$$

Recall dispersion $\sigma(rr)_{i,t}$ is the weighted standard deviation of recalled revisions for firm i in month t , defined in (3.10). Disagreement_{*i,t*} is the cross-sectional standard deviation of individual analyst EPS forecasts scaled by last month's stock price, following Diether et al. (2002). AbVol_{*i,t*} is abnormal log trading volume for firm i in month t , following Cookson and Niessner (2020), which equals log volume in month t minus the average log volume from months $t-12$ to $t-2$. The control variables $Z_{i,t}$ includes lagged monthly returns $Ret_{i,t-1}$, $Ret_{i,t-2}$, and $Ret_{i,t-3}$, and return volatility $\sigma(Ret)_{i,t-1}$, measured as the standard deviation of monthly returns over months $t-12$ through $t-1$. All specifications include firm and time fixed effects γ_i and η_t . Coefficients on recall dispersion $\sigma(rr)_{i,t}$ are scaled by 1,000 for readability. The sample period is January 2007 to December 2020. Standard errors are two-way clustered by industry and year and are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table V: Analyst Forecast Errors and Recall Distortion

	Forecast error $e_{i,t}$		Forecast error $ e_{i,t} $		Forecast direction error $e_{i,t}^d$	
	(1)	(2)	(3)	(4)	(5)	(6)
	Linear	Linear	Linear	Linear	Ologit	Ologit
$\Delta R_{i,t}$	0.041*** (0.009)	0.039*** (0.009)			0.024*** (0.003)	0.023*** (0.003)
$ \Delta R_{i,t} $			0.125*** (0.025)	0.094*** (0.019)		
Lagged forecast revision		0.008 (0.019)		0.032** (0.014)		-0.020** (0.008)
Lagged earnings growth		-0.004 (0.002)		0.010** (0.003)		-0.002 (0.001)
Lagged forecast		-1.255** (0.501)		2.679*** (0.543)		-0.389* (0.226)
Firm fixed effect	Yes	Yes	Yes	Yes	No	No
Month fixed effects	Yes	Yes	Yes	Yes	No	No
Observations	277,410	277,410	277,410	277,410	277,487	277,487
R-squared	0.139	0.145	0.267	0.301		

This table reports regression results for analyst forecast errors and recall distortion. Columns (1)–(2) estimate

$$e_{i,t} = \beta \times \Delta R_{i,t} + \theta \times Z_{i,t} + \gamma_i + \eta_t + \varepsilon_{i,t}.$$

Columns (3)–(4) estimate

$$|e_{i,t}| = \beta \times |\Delta R_{i,t}| + \theta \times Z_{i,t} + \gamma_i + \eta_t + \varepsilon_{i,t}.$$

Columns (5)–(6) estimate an ordered logit for the direction of the forecast error,

$$\Pr(e_{i,t}^d = j) = \Lambda(\kappa_j - x_{i,t}) - \Lambda(\kappa_{j-1} - x_{i,t}), \quad j \in \{-2, -1, 0, 1, 2\},$$

where $x_{i,t} = \beta \times \Delta R_{i,t} + \theta \times Z_{i,t} + \gamma_i + \eta_t$ and $\Lambda(z) = 1/(1 + \exp(-z))$. The cutoffs κ_j are estimated jointly with the coefficients. The dependent variables are the analyst forecast error, $e_{i,t}$, the magnitude of analyst forecast error $|e_{i,t}|$, and the error direction $e_{i,t}^d \in \{-2, -1, 0, 1, 2\}$, for firm i at time t . $\Delta R_{i,t}$ is the analyst recall distortion defined in (4.1). Columns (2), (4), and (6) add three controls: (i) lagged forecast revision $(F_{i,t-1}[EPS_{i,t+l}] - F_{i,t-2}[EPS_{i,t+l}])/P_{i,t-1}$; (ii) lagged earnings growth $(EPS_{i,t-l_1} - EPS_{i,t-l_2})/P_{i,t-1}$; and (iii) the lagged forecast $F_{i,t-1}[EPS_{i,t+l}]/P_{i,t-1}$, where $F_{i,t}[EPS_{i,t+l}]$ is the consensus analyst forecast for firm i 's future EPS at time t , $EPS_{i,t-l_1}$ and $EPS_{i,t-l_2}$ are realized earnings over the prior two fiscal years, and $P_{i,t-1}$ is the lagged stock price. The sample period is January 2007 to December 2020. Standard errors are two-way clustered by industry and year and are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table VI: Recall Distortions and Forecast Revisions

Month	1	2	3	4	5	6
Mean	-18.12	-2.42	2.69	-3.81	-1.38	1.08
	(-6.73)	(-1.52)	(1.54)	(-1.12)	(1.20)	(0.57)

This table reports the difference in average analyst forecast revisions (ΔF) between the fifth and first quintiles of portfolios sorted by recall distortions (ΔR , defined in (4.1)). The first quintile includes stocks with overly positive analyst recalls, and the fifth quintile includes stocks with overly negative recalls. Columns (1) to (6) report the difference one to six months after portfolio formation. The sample period is January 2007 to December 2020. The t -statistics reported in parentheses are robust to heteroskedasticity and autocorrelation with up to 12 lags.

Table VII: Biased Recall of the Prior Month and Short-Term Reversal

$\Delta\rho$	Short-term reversal			
	1 (Low)	2	3 (High)	3 – 1
1 (Over-recall)	0.80 (3.11)	0.69 (4.15)	0.22 (0.92)	-0.58 (-1.79)
2	0.39 (3.70)	0.45 (5.34)	0.73 (6.04)	0.34 (2.39)
3 (Under-recall)	0.35 (2.16)	0.38 (3.33)	0.86 (5.04)	0.51 (2.11)
3 – 1	-0.44 (-1.56)	-0.32 (-1.96)	0.65 (2.96)	1.09 (3.88)

This table reports average risk-adjusted returns for equal-weighted portfolios that are double-sorted first by biased investor recall of the prior month ($\Delta\rho$, defined in (4.6)), and then by prior-month returns. Risk-adjusted returns, reported in percent, are estimated using the Carhart four-factor model (Carhart, 1997). The sample period is January 2007 to December 2020. The t -statistics reported in parentheses are robust to heteroskedasticity and autocorrelation with up to 12 lags.

Table VIII: Misreactions in Perceptions, Forecast Misreactions, and Biased Recency Effect

	Misreaction	Biased Recency Effect	
		Normal	Crisis
	(1)	(2)	(3)
$\Delta^2\omega$	-0.010** (0.005)	0.017** (0.008)	-0.018*** (0.004)
Observations	180	144	36
R-squared	0.028	0.039	0.223

This table reports OLS regressions of forecast misreaction and biased recency effect on misreaction in the analyst's perceived changes in external conditions ($\Delta^2\omega_t$, defined in (5.1)). Forecast misreaction (ΔReact_t) and biased recency effect ($\Delta\text{Recency}_t$) are defined in (5.3) and (5.5). The sample period is January 2007 to December 2020 with year 2008, 2009, and 2020 are classified as crisis periods, while the remaining years are considered normal periods. The sample period is January 2007 to December 2020. Standard errors are two-way clustered by industry and year and are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table IX: Paired t -statistics Between LSTM Models with and without Selective Forgetting

Model	Recency Effect (1)	Recalled Revisions (2)
<i>Panel A: Analyst</i>		
LSTM with forget gate	0.052	-0.707
LSTM without forget gate	0.104	-0.625
Difference	0.051 (4.75)	0.082 (6.69)
<i>Panel B: Benchmark</i>		
LSTM with forget gate	0.177	-0.434
LSTM without forget gate	0.160	-0.488
Difference	-0.017 (-4.72)	-0.054 (-4.27)

This table reports paired t -statistics comparing the full LSTM model with a counterfactual LSTM without the forget gate. Panel A presents results for the analyst model, and Panel B presents results for the benchmark model. Column (1) reports paired t -statistics for the recency effect (Recency, defined in (5.4)). Column (2) reports paired t -statistics for recalled revisions (RR , defined in (3.6)). The sample period is January 2007 to December 2020. The t -statistics reported in parentheses are two-way clustered by industry and year.

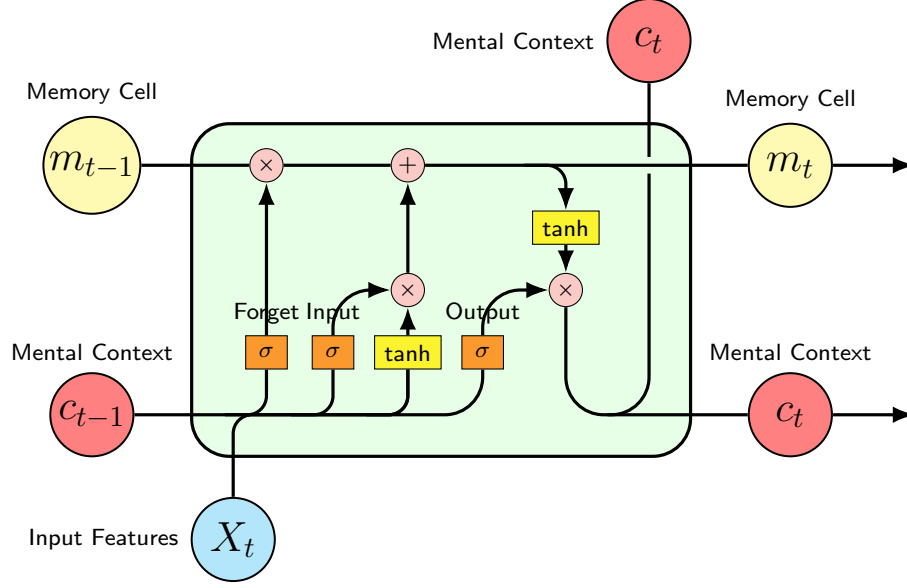


Figure I: The Structure of an LSTM Unit

The figure displays the structure of an LSTM unit. The inputs of an LSTM unit are the prior mental context c_{t-1} , the prior memory cell m_{t-1} , and the current features X_t ; the outputs are the updated memory cell m_t and the current mental context c_t . Information flow is regulated by the forget, input, and output gates, see Equations (2.1) to (2.6). “ $\times$ ” and “ $+$ ” denote element-wise multiplication and addition. σ is the sigmoid function and $\tanh$ is the hyperbolic tangent.

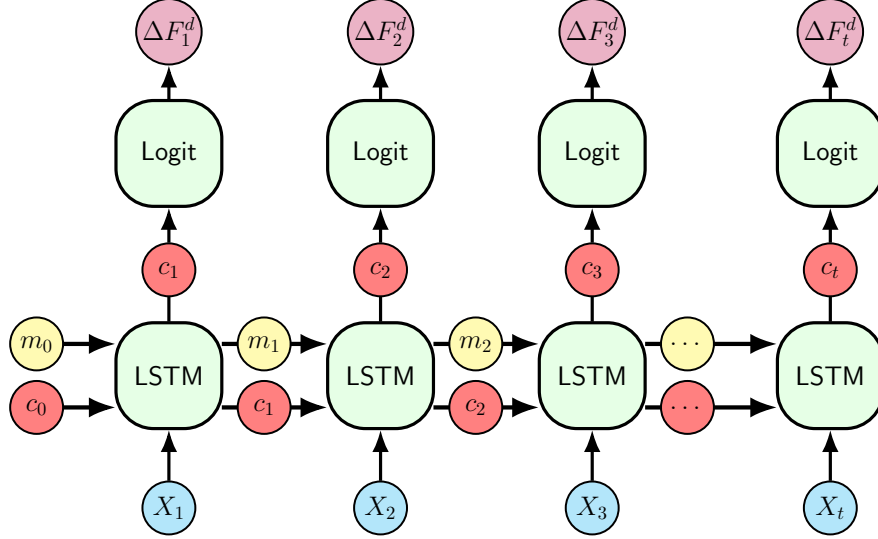


Figure II: Recursive LSTM Architecture for Analyst Forecast Revisions

The figure shows the recursive LSTM sequence used to model analyst forecast revisions over time. For simplicity, the firm subscript i is omitted in the diagram, so $X_{i,t}$ appears as X_t . At date t , the cell takes external features $X_{i,t}$ and the prior states $(m_{i,t-1}, c_{i,t-1})$, updates them to the current memory $m_{i,t}$ and mental context $c_{i,t}$, and passes these states forward. The internal structure of an LSTM unit is illustrated in Figure I. The mental context $c_{i,t}$ is mapped to the direction of the forecast revision ($\Delta F_{i,t}^d$, defined in (2.7)) via a multinomial logit function. Time flows from left to right.

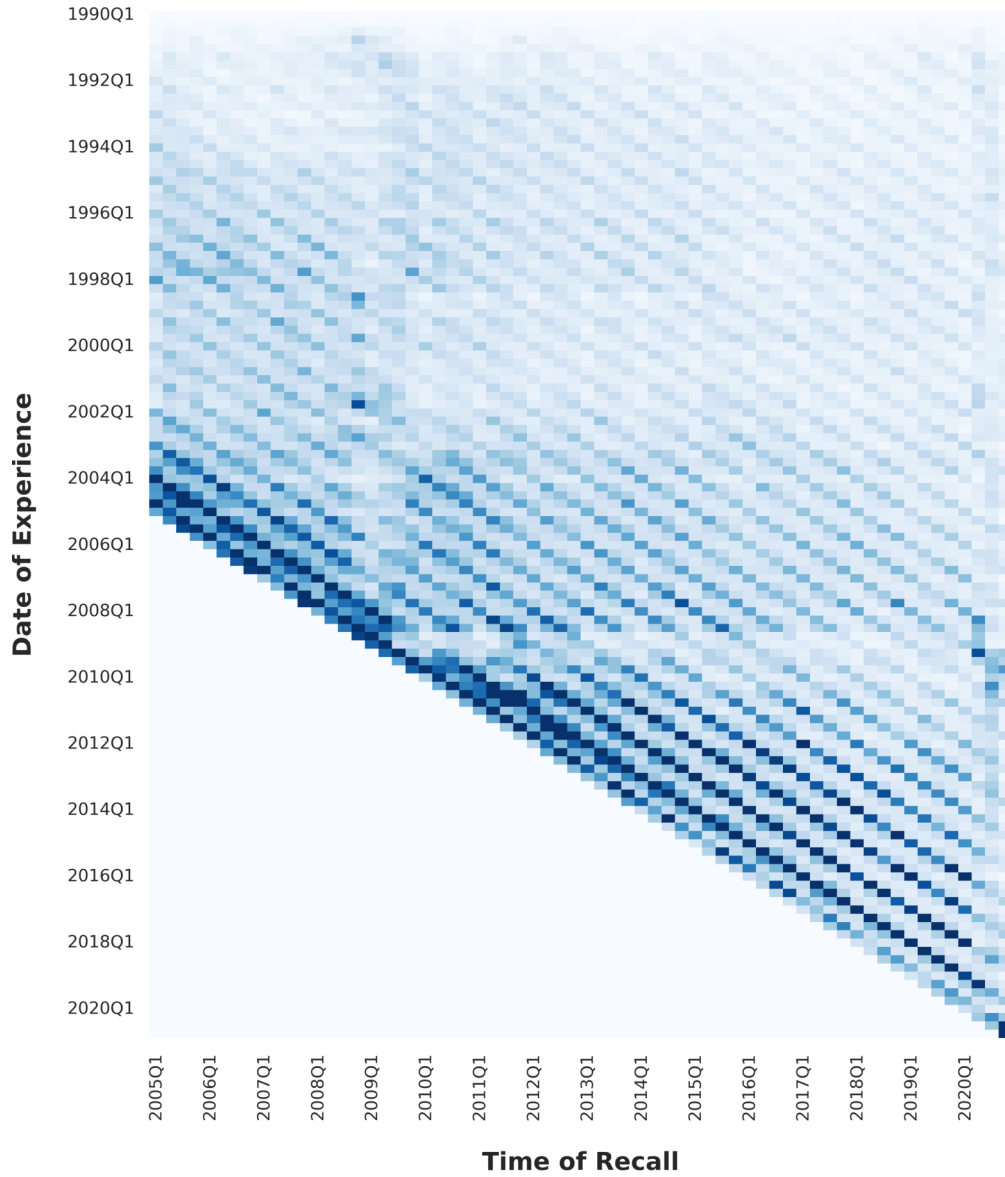


Figure III: Analyst LSTM Recalls

The heatmap shows how often past experiences from date y (rows) are recalled at the current time x (columns) from the analyst LSTM model. For each firm at time x , we compute the similarity between the current mental context and all past mental contexts estimated by the model, select the five most similar past experiences, and record their occurrence dates across all firms. The color intensity indicates the percentage of recalls from each date y , with the scale capped at 5% for display clarity. Darker blue areas indicate that experiences from date y are more frequently recalled at time x .

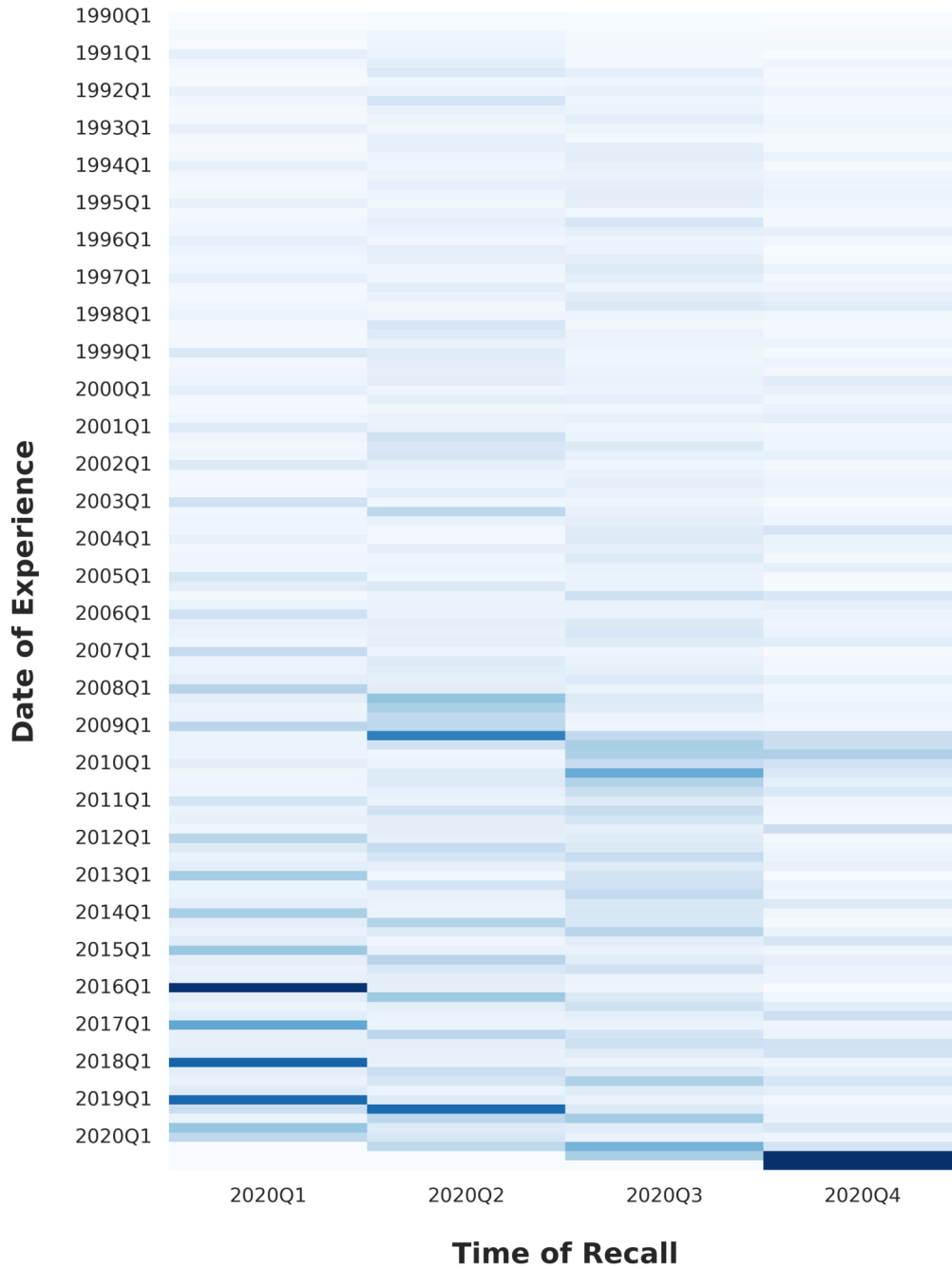


Figure IV: Analyst LSTM Recalls during COVID

The heatmap shows, for 2020, how often past experiences from date y (rows) are recalled at the current time x (columns) from the analyst LSTM model. For each firm at time x , we compute the similarity between the current mental context and all past mental contexts estimated by the model, select the five most similar past experiences, and record their occurrence dates across all firms. The color intensity indicates the percentage of recalls from each date y , with the scale capped at 5% for display clarity. Darker blue areas indicate that experiences from date y are more frequently recalled at time x .

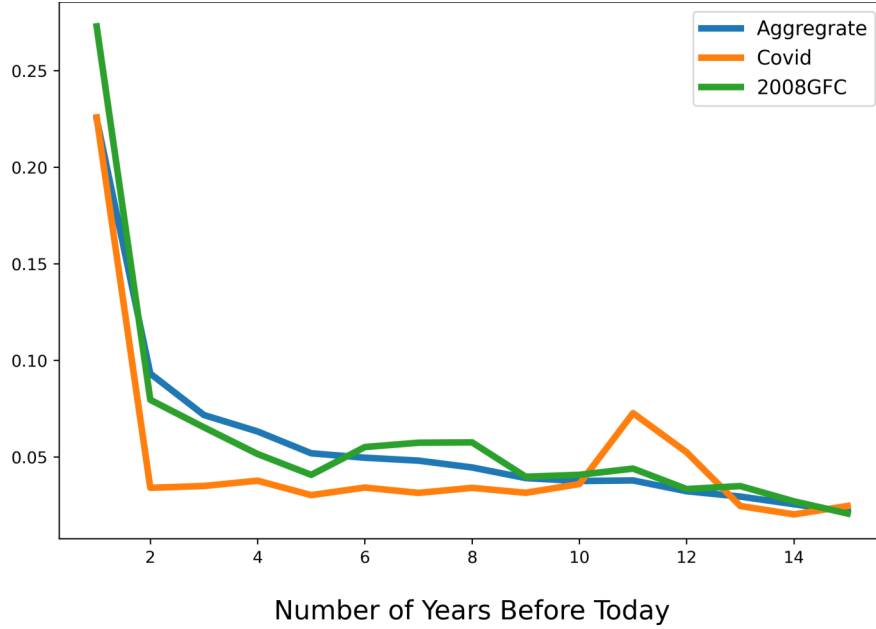


Figure V: Distribution of Time Lags between Recall Dates and Recalled Episodes

This figure plots the frequency with which recalled episodes occurred a given number of years before the recall date. For each firm at each time, we identify the five most similar past experiences based on the similarity between current and past mental contexts estimated by the analyst LSTM model. We then compute the distribution of time lags (in years) between recall dates and recalled episodes. The figure reports the average distribution over 2005–2020 (“Aggregate”) and separately for periods around the 2008 Global Financial Crisis (“2008GFC”) and the COVID pandemic (“Covid”).

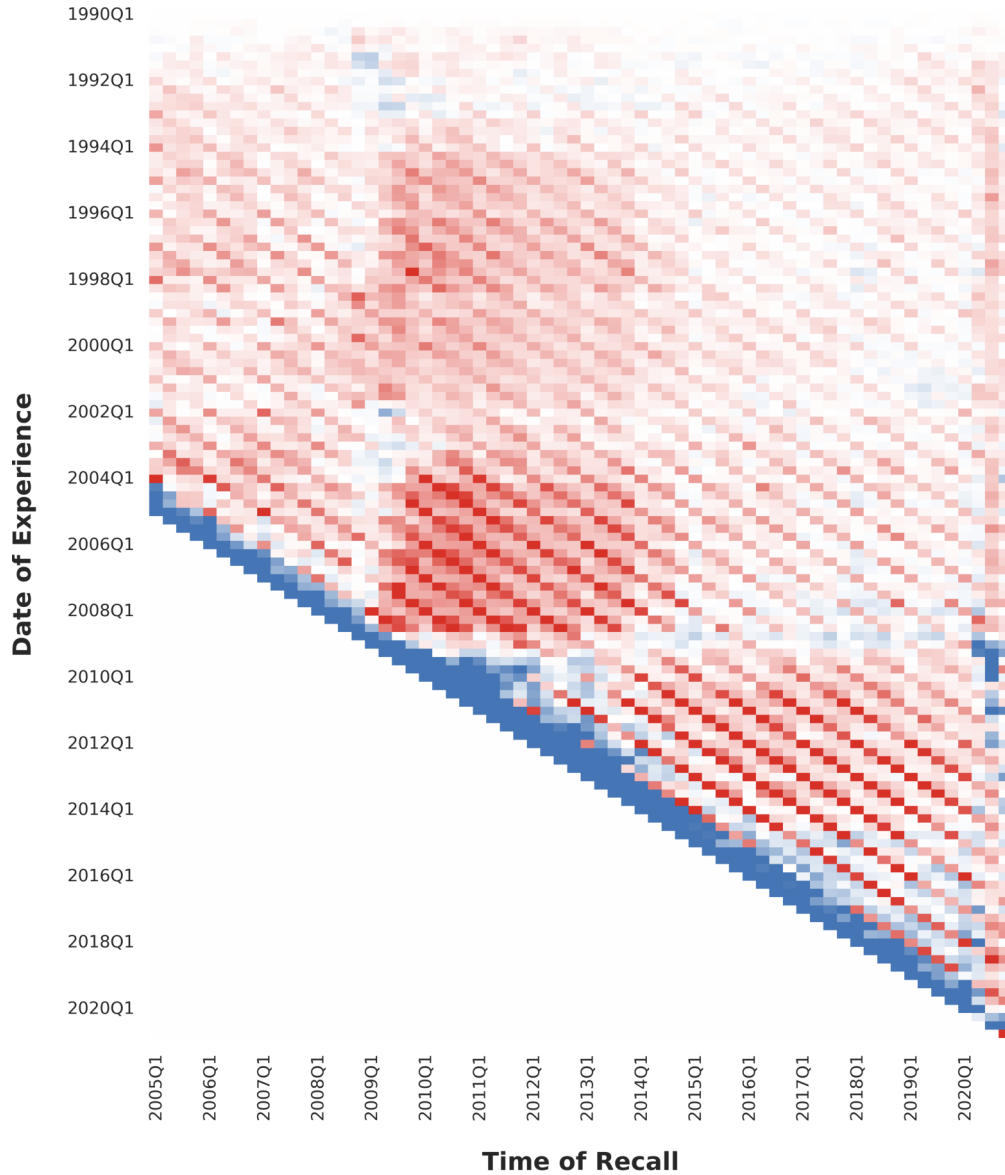


Figure VI: Difference between Analyst LSTM Recalls and Benchmark LSTM Recalls

This heatmap shows differences in recall intensity between the benchmark LSTM model and the analyst LSTM model. Rows index the date of the recalled experience y , and columns index the recall date x . For each model, firm, and recall date x , we compute the similarity between the current mental context and all past mental contexts estimated by that model, select the five most similar past experiences, and record their occurrence dates across firms. We then subtract the benchmark recall percentage from the analyst recall percentage for each pair of dates (x, y) . The color intensity indicates the difference in percentage of recalls from each date y , with a scale capped at $\pm 2.5\%$ for display clarity. Darker red areas indicate that the analyst recalls experiences from date y more frequently than the benchmark, while darker blue areas indicate the opposite.

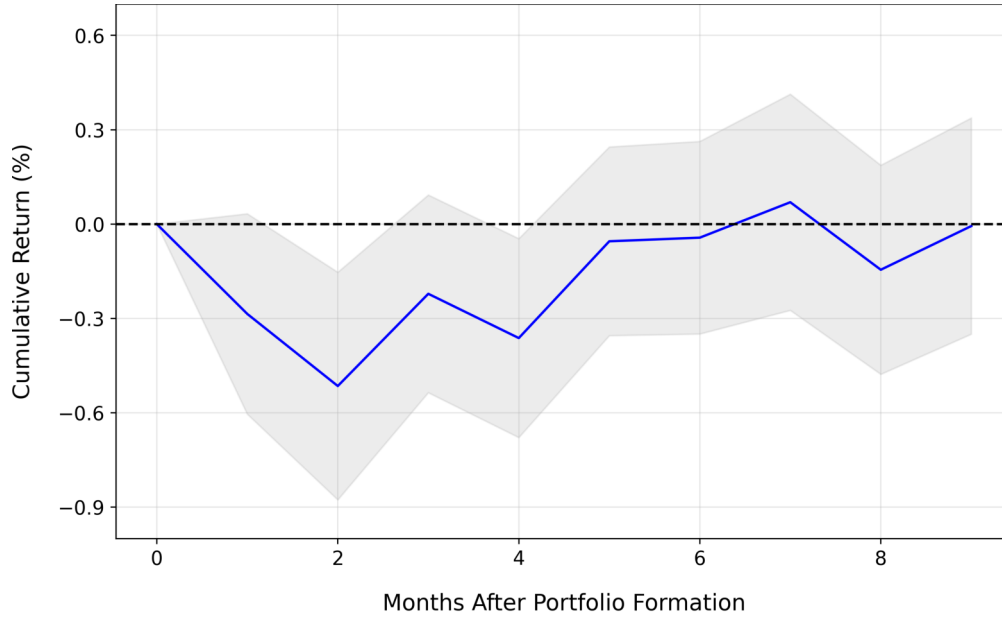


Figure VII: Cumulative Returns on Long-Short Portfolio Sorted by Recall Distortion

This figure plots the cumulative returns of a value-weighted long-short strategy sorted by recall distortion (ΔR , defined in (4.1)). The portfolio goes long the quintile of stocks with the most negative recall distortion and shorts the quintile with the most positive recall distortion. The shaded area shows 95% confidence intervals. The sample period is January 2007 to December 2020.

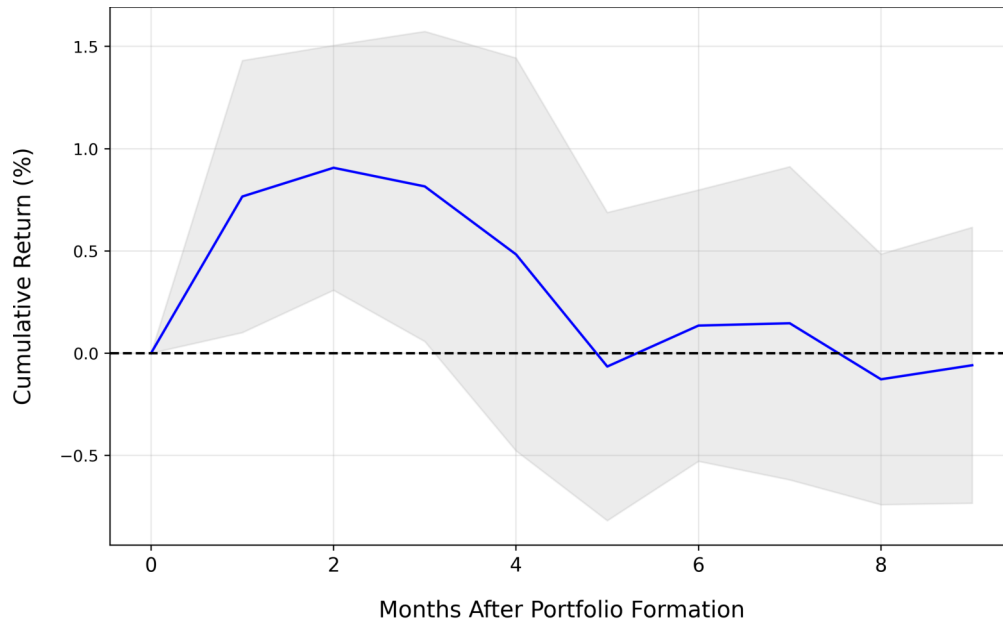
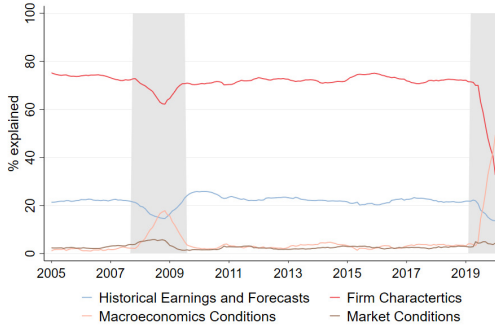
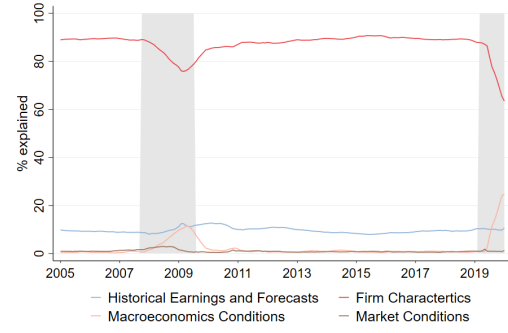


Figure VIII: Cumulative Returns of Double-Sorted Portfolio by Biased Recall and Short-Term Reversal

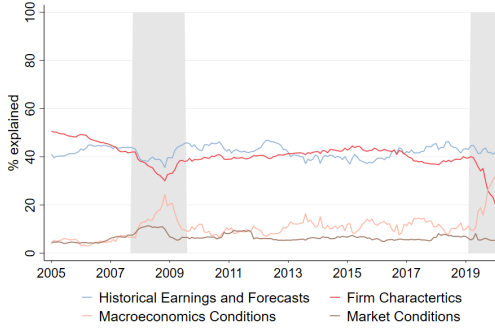
This figure shows the cumulative returns of a monthly long-short strategy combining biased recall of the prior month with short-term reversal. Each month, stocks are first sorted into terciles by biased recall ($\Delta\rho$ in (4.6)). Within each $\Delta\rho$ tercile, firms are sorted by prior-month returns, and a reversal portfolio is formed as past losers minus past winners. The strategy goes long the reversal portfolio in the under-recall tercile and shorts it in the over-recall tercile. Shaded areas indicate 95% confidence intervals. The sample period is January 2007 to December 2020.



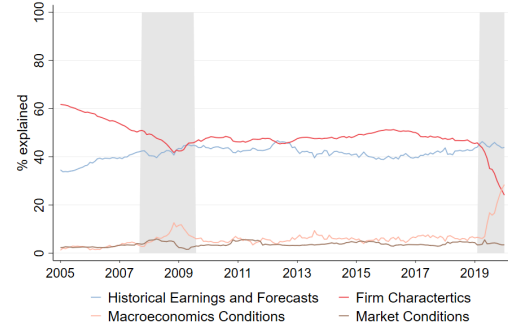
(a) Benchmark - Mental Context



(b) Benchmark - Memory Cell



(c) Analyst - Mental Context



(d) Analyst - Memory Cell

Figure IX: Variable Importance Decomposition of Mental Contexts and Memory Cells

This figure shows the time series of group variable importance for the benchmark and analyst LSTM models. Panels display (a) benchmark mental context, (b) benchmark memory cells, (c) analyst mental context, and (d) analyst memory cells. We aggregate the 79 features into four groups and sum variable importance for feature j at time t ($\omega_{j,t}$) within each group, corresponding to Table A.1: (i) ω_t^C , firm characteristics excluding earnings-related variables; (ii) ω_t^M , macroeconomic conditions; (iii) ω_t^R , market conditions; (iv) ω_t^F , historical earnings and forecasts. $\omega_{j,t}$ is computed following Gu et al. (2020): for each LSTM dimension, we regress it on all external features using a 12-month rolling window, then set feature j to zero while keeping others fixed and measure the resulting reduction in R^2 as $\omega_{j,t}$.

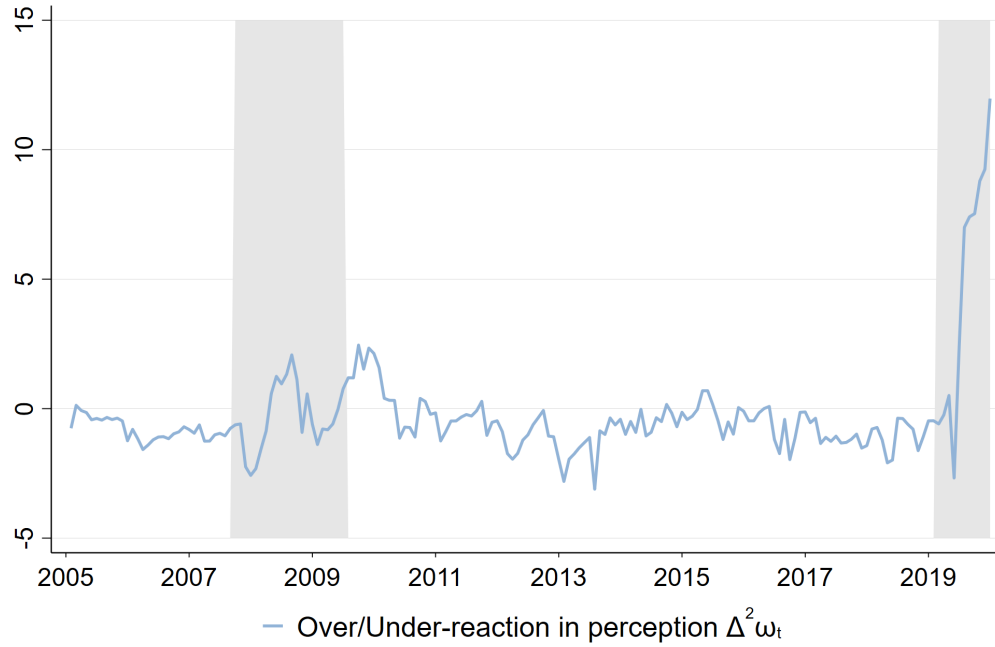


Figure X: Misreaction in Perceived External Economic Conditions

This figure plots the time series of analyst misreaction in perceptions, $\Delta^2\omega_t$ (defined in (5.2)), which measures deviations of the analyst perception of environmental changes ($\Delta\omega_t$, defined in (5.1)), from the benchmark. Positive values ($\Delta^2\omega_t > 0$) indicate underreaction, while negative values ($\Delta^2\omega_t < 0$) indicate overreaction.

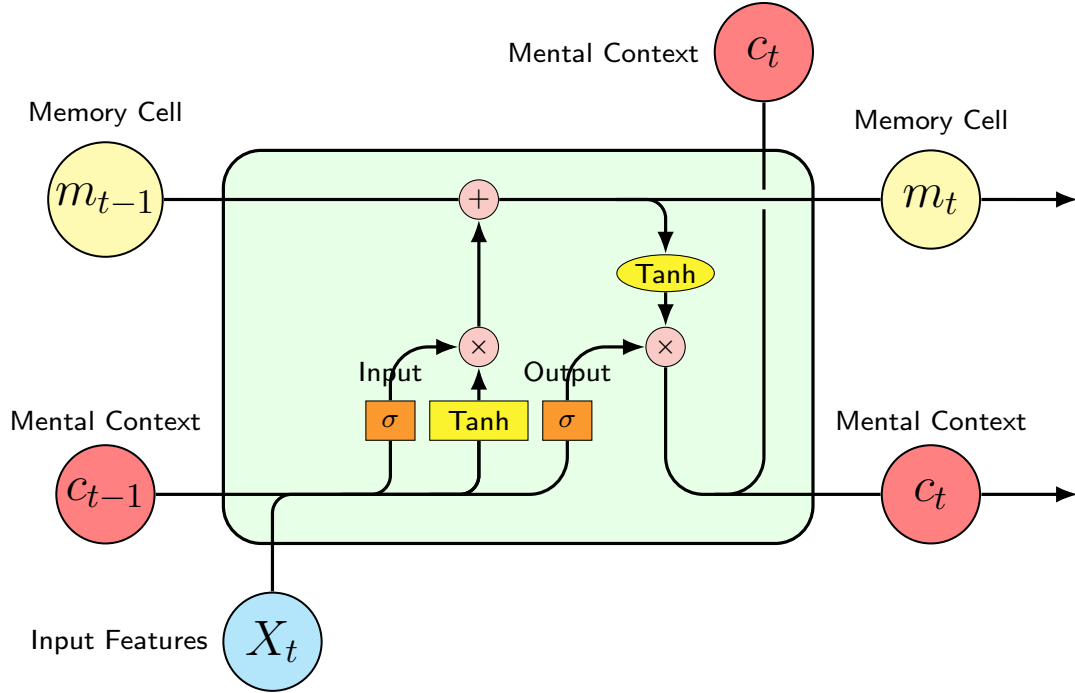


Figure XI: The Structure of an LSTM Unit without Forget Gate

This figure shows an LSTM unit without the forget gate. We set $\text{forget} \equiv 1$ so that the memory cell updates without selective forgetting, $m_t = m_{t-1} + \text{input}_t \times \tilde{x}_t$. All other components and parameters keep fixed as illustrated in Figure I.

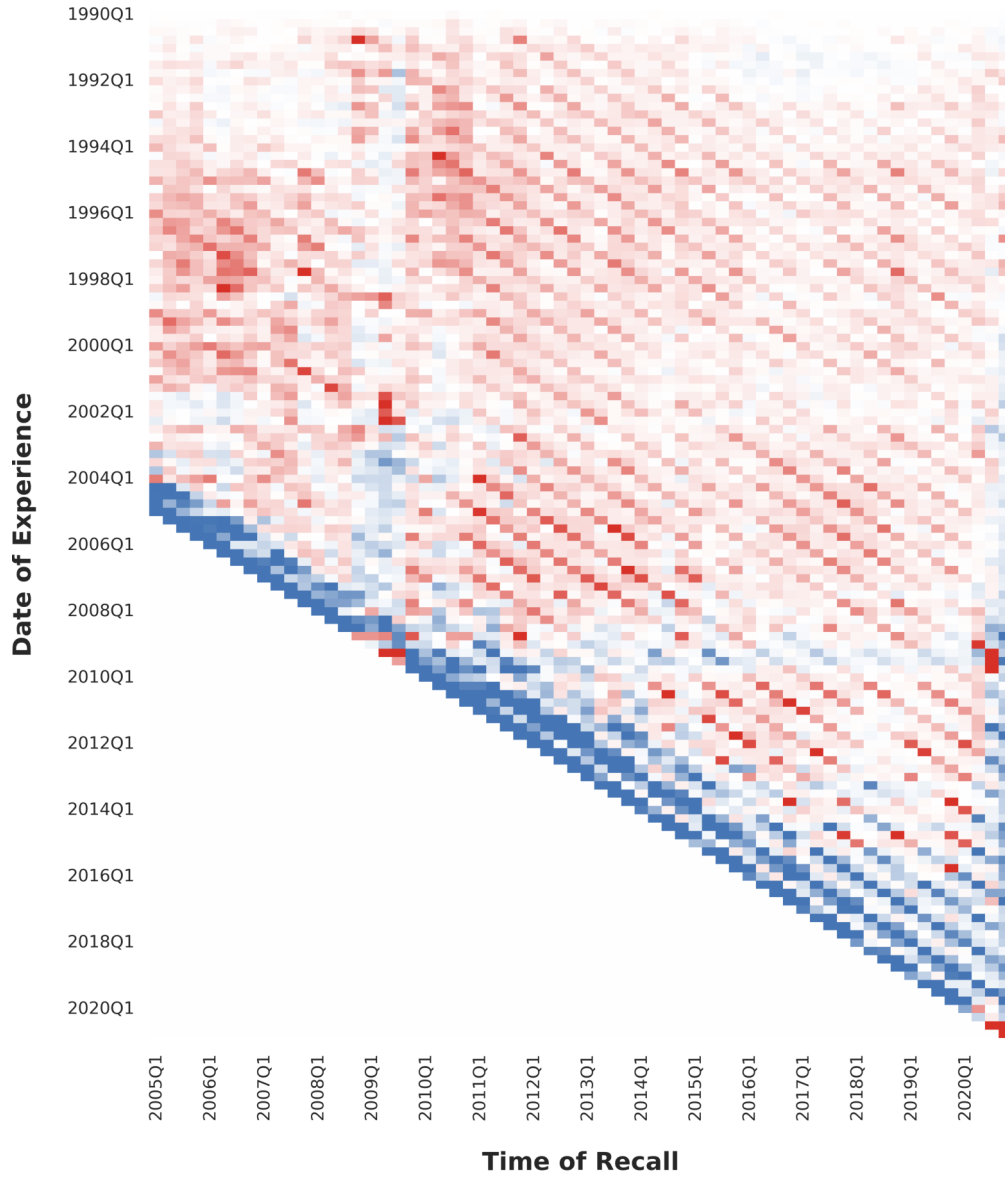


Figure XII: Difference between Analyst LSTM Recalls and Counterfactual Recalls

This heatmap shows differences in recall intensity between the analyst LSTM model and counterfactual LSTM model (without forget gate). Rows index the date of the recalled experience y , and columns index the recall date x . For each model, firm, and recall date x , we compute the similarity between the current mental context and all past mental contexts estimated by that model, select the five most similar past experiences, and record their occurrence dates across firms. We then subtract the counterfactual recall percentage from the LSTM recall percentage for each pair of dates (x, y) . The color scale is capped at $\pm 2.5\%$ for display clarity. Darker red areas indicate that the analyst recalls experiences from date y more frequently with selective forgetting than without it, while darker blue areas indicate the opposite.

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Appendix

Appendix A.1 External Features

Table A.1 defines the 79 external features X . Variable construction follows van Binsbergen et al. (2023). We assign variables into four groups: firm characteristics, macroeconomic conditions, market conditions, and historical earnings and forecasts.³⁸

Sources by group: (i) Firm characteristics: all variables are from the WRDS Financial Ratios,³⁹ except for *asset_g*, *invest_g*, and *sales_g*, which are from Compustat; (ii) Macroeconomic conditions: all variables are from Federal Reserve Bank of Philadelphia; (iii) Market conditions: the monthly stock return is from CRSP; (iv) Historical earnings and forecasts: *capei*, *pe_exi*, *pe_inc*, and *PEG_trailing* are from WRDS Financial Ratios; all remaining variables in this category are from I/B/E/S.

Table A.1: Definition of External Features

(i) Firm Characteristics			
Variable	Definition	Variable	Definition
Accrual	Accruals/Average Assets	adv_sale	Advertising Expenses/Sales
aftret_eq	After-tax Return on Average Common Equity	aftret_equity	After-tax Return on Total Stockholders Equity
aftret_invcapx	After-tax Return on Invested Capital	asset_g	Growth Rate in Total Assets
at_turn	Asset turnover	bm	Book/Market
capital_ratio	Capitalization Ratio	cash_debt	Cash Flow/Total Debt
cash_lt	Cash Balance/Total Liabilities	cash_ratio	Cash Ratio
cfm	Cash Flow Margin	curr_debt	Current Liabilities/Total Liabilities
curr_ratio	Current Ratio	debt_asset	Total Debt/Total Assets
debt_at	Total Debt/Total Assets	debt_capital	Total Debt/Capital
debt_ebitda	Total Debt/EBITDA	debt_invcap	Long-term Debt/Invested Capital
divyield	Dividend Yield	dltt_be	Long-term Debt/Book Equity
dpr	Dividend Payout Ratio	efftax	Effective Tax Rate

³⁸We exclude earnings-related variables, i.e., *capei*, *pe_exi*, *pe_inc*, and *PEG_trailing*, from this category because they are tied to historical earnings. These variables are placed in the “historical earnings and forecasts” category. We use this grouping in Section 5.1.

³⁹See <https://wrds-www.wharton.upenn.edu/pages/grid-items/financial-ratios-firm-level/>

equity_invcap	Common Equity/Invested Capital	evm	Enterprise Value Multiple
fcf_ocf	Free Cash Flow/Operating Cash Flow	gpm	Gross Profit Margin
GProf	Gross Profit/Total Assets	int_debt	Interest/Average Long-term Debt
int_totdebt	Interest/Average Total Debt	intcov	After-tax Interest Coverage
intcov_ratio	Interest Coverage Ratio	invest_g	Growth Rate in Capital Expenditure
inv_turn	Inventory Turnover	invt_act	Inventory/Current Assets
lt_ppent	Total Liabilities/Total Tangible Assets	npm	Net Profit Margin
ocf_lct	Operating CF/Current Liabilities	opmad	Operating Profit Margin After Depreciation
opmbd	Operating Profit Margin Before Depreciation	pay_turn	Payables Turnover
pcf	Price/Cash flow	pretret_earnat	Pre-tax Return on Total Earning Assets
pretret_noa	Pre-tax return on Net Operating Assets	profit_lct	Profit Before Depreciation/Current Liabilities
ps	Price/Sales	ptb	Price/Book
ptpm	Pre-tax Profit Margin	quick_ratio	Quick Ratio (Acid Test)
RD_SALE	Research and Development/Sales	rect_act	Receivables/Current Assets
rect_turn	Receivables Turnover	roa	Return on Assets
roce	Return on Capital Employed	roe	Return on Equity
sale_equity	Sales/Stockholders Equity	sale_invcap	Sales/Invested Capital
sale_nwc	Sales/Working Capital	sales_g	Growth Rate in Sales
short_debt	Short-Term Debt/Total Debt	totdebt_invcap	Total Debt/Invested Capital
(ii) Macroeconomic Conditions			
Variable	Definition	Variable	Definition
con_g	Log Difference of Consumption in Goods and Services	GDP_g	Log Difference of Real GDP
IPT_g	Log Difference of Industrial Production Index	unemployment	Unemployment Rate
(iii) Market Conditions			
Variable	Definition		
return	Monthly Stock Return		
(iv) Historical Earnings and Forecasts			
Variable	Definition	Variable	Definition

AF_EP_lag	Mean Analyst Forecast from Last Period /Stock Price from Last Month	AF_g_lag	Lag 1 Growth Rate in Mean Analyst Forecast
capei	Shillers Cyclically Adjusted P/E ratio	Maturity	Months to Fiscal End Date/12
NUMEST_lag	Number of Forecasts from Last Period	pe_exi	P/E (Diluted, Excl. EI)
pe_inc	P/E (Diluted, Incl. EI)	PEG_trailing	Trailing P/E to Growth ratio
Realized_ANN_g	Growth Rate in Realized Annual Earnings	Realized_EP_ANN	Realized Annual Earnings from Last Period/Stock Price from Last Month
Realized_EP_QTR	Realized Quarter Earnings from Last Period/Stock Price from Last Month	Realized_QTR_g	Growth Rate in Realized Quarter Earnings

Appendix A.2 Model Training and Tuning

In this appendix, we describe the procedures for model training, sample splitting, and performance evaluation, following Gu et al. (2020). The base training sample covers January 1990 to December 2004, and the validation sample covers January 2005 to December 2006.⁴⁰ The validation sample is used to select the hyperparameter K , the dimension of the memory cell and mental context. The test sample spans January 2007 to December 2020.

We train the model using the Adam optimizer (Kingma and Ba, 2014) with default settings. To prevent overfitting, we apply early stopping with a patience of 5. The batch size is 10,000, and the loss function is the negative log-likelihood, consistent with the multinomial logit output.

The only hyperparameter requiring tuning is K , the dimension of memory cells and mental contexts. We evaluate candidate values of K by training on the base sample and assessing performance on the validation sample to select the optimal value.

Table A.2: Model Performance across Hyperparameter K

Dimension K	5	10	15	20
<i>Panel A: Analyst Forecast Revisions (ΔF^d)</i>				
Training	59.58%	61.42%	62.26%	62.04%
Validation	54.22%	55.45%	55.15%	54.72%
<i>Panel B: Realized Earnings Revisions (ΔRE^d)</i>				
Training	68.68%	71.22%	72.33%	71.74%
Validation	59.58%	58.95%	57.73%	57.58%

This table presents the model performance for fitting and predicting the direction of analyst forecast revision (ΔF^d , see (2.7)) and realized earnings revision (ΔRE^d , see (2.8)) in the base training and validation samples with different choices of dimension of the memory cells and mental contexts (K). The training sample is from January 1990 to December 2004. The validation sample is from January 2005 to December 2006.

Table A.2 reports the model’s performance in fitting and predicting the direction of analyst forecast revisions (ΔF^d , (2.7)) and realized earnings revisions (ΔRE^d , (2.8)) using different dimensions of the memory cells and mental contexts (K). Based on validation

⁴⁰All variables are standardized following common practice: each variable is centered and scaled using the mean and standard deviation from the base training sample. The same parameters are applied to the validation and test samples.

accuracy, $K = 10$ is optimal for ΔF^d and $K = 5$ for ΔRE^d , though differences are small. For consistency, we use $K = 10$ for both models throughout the paper.

Beyond the base training sample, we employ a recursive (expanding window) scheme to train the model on subsequent periods and evaluate its out-of-sample performance. After selecting the optimal hyperparameter K , we first train the model on data from January 1990 to December 2004 and conduct out-of-sample predictions for 2005. We then expand the training sample to include data from 2005, retrain the model starting from the previous parameters, and evaluate predictions for 2006. This process is repeated annually through 2020.

This recursive approach provides two main advantages. First, it allows the model to adapt to changes in the economic and financial environment, improving predictive accuracy over time. Second, it mirrors analyst learning by updating beliefs as new information becomes available, so that perceptions of a firm’s past state can evolve as episodes are revisited with additional experience.

Appendix A.3 Temporal Contiguity

In this appendix, we conduct a simulation study to examine the temporal contiguity effect in the LSTM for analyst belief formation.

Temporal contiguity refers to the tendency for recalling one episode to increase the likelihood of recalling nearby episodes in time. Laboratory evidence (Kahana, 1996) shows that after recalling an item at a given serial position, subjects are more likely to recall items from neighboring positions. Temporal contiguity has two main properties: forward asymmetry, meaning recalls tend to move toward later items rather than earlier ones, and time-scale invariance, meaning the effect remains strong even for items from the distant past. Time-scale invariance motivates the inclusion of long-term memory in our setting. Beyond psychology, Wachter and Kahana (2024) discuss temporal contiguity in a theoretical finance framework, and we provide empirical evidence for it in analyst belief formation.

LSTMs naturally combine long-term memory with an autoregressive mental context, allowing them to reproduce temporal contiguity (Howard and Kahana, 2002). However, empirically demonstrating this effect in financial settings is challenging. Economic and financial variables are typically persistent, so episodes that are close in time tend to have similar external features. As a result, recalling one episode may mechanically increase the likelihood of recalling its neighbors even without any memory effect. Temporal contiguity, by contrast, implies that even when two episodes are unrelated in content, recalling one increases the likelihood of recalling the other simply because they occurred close together in time.

To isolate this effect from the mechanical similarity induced by persistence, we design a simulation that removes autocorrelation in the external features by rotating the inputs to an orthogonal basis. The simulation follows the recursive procedure described in Section 2.3 and uses the parameters estimated from the analyst LSTM model for ΔF^d .

The simulation proceeds as follows. For each round, we generate simulated inputs:

$$X_t^{\text{sim}} = \mathbb{E}[X] + 10u_t \times \sigma(X), \quad t = 1, 2, \dots, T,$$

where $\mathbb{E}[X]$ and $\sigma(X)$ are the time-series mean and element-wise standard deviations of the original features X , and “ $\times$ ” denotes element-wise multiplication. The vectors u_t are randomly drawn, mutually orthogonal, and have L^2 norm of one. By construction, the inner product between any two simulated inputs X_t^{sim} equals the $\|\mathbb{E}[X]\|^2$, which effectively

removes autocorrelation while keeping inputs on a comparable empirical scale (the factor 10 serves this purpose).⁴¹ We generate $T = 70$ periods of simulated inputs.⁴²

In period $T + 1$, we duplicate a randomly selected input X_τ^{sim} , setting $X_{T+1}^{\text{sim}} = X_\tau^{\text{sim}}$, where $\tau \in \{11, \dots, T - 10\}$ to limit primacy and recency effects. Using the empirically estimated analyst LSTM model, we generate mental contexts $\{c_t^{\text{sim}}\}_{t=1}^{T+1}$ from the simulated inputs $\{X_t^{\text{sim}}\}_{t=1}^{T+1}$ and use c_{T+1}^{sim} as the retrieval cue.

Temporal contiguity is assessed by the similarity between c_{T+1}^{sim} and other mental contexts:

$$S(c_{\tau+\ell}^{\text{sim}}, c_{T+1}^{\text{sim}}),$$

defined in (2.9), which represents the likelihood that episode at $\tau + \ell$ is recalled when cued by c_{T+1}^{sim} .⁴³

If temporal contiguity exists, similarity is expected to decrease as the distance from τ increases, with a forward asymmetry. Formally, we expect:

$$\begin{aligned} S(c_{\tau+i}, c_{T+1}) &> S(c_{\tau+j}, c_{T+1}), \quad \forall 0 < i < j; \\ S(c_{\tau-i}, c_{T+1}) &> S(c_{\tau-j}, c_{T+1}), \quad \forall 0 < i < j; \\ S(c_{\tau+i}, c_{T+1}) &> S(c_{\tau-i}, c_{T+1}), \quad \forall i > 0. \end{aligned} \tag{6.1}$$

Figure A.1 shows the estimated average temporal contiguity effect. The x -axis represents the lag ℓ relative to τ . The y -axis shows similarity $S(c_{\tau+\ell}^{\text{sim}}, c_{T+1}^{\text{sim}})$ for lags $\ell = -10, \dots, 10$, normalized are normalized by the minimum and maximum across $\ell \neq 0$, so that the highest (lowest) value in the plotted window is 1 (0). The simulation is repeated 10,000 times.

Subfigure A.1a presents temporal contiguity effect of the trained analyst LSTM. Similarity is highest immediately after τ and declines as the lag increases. For negative lags, similarity peaks just before τ and decreases as $|l|$ grows. For equal absolute lags $|\ell|$, similarity is generally lower for negative lags than for positive lags. Overall, this pattern aligns with the conditions for temporal contiguity in (6.1).

However, these conditions are not automatically satisfied by the LSTM structure. We show two examples. In Subfigure A.1b, we keep all other parameters fixed while setting

⁴¹Our results are robust to the choice of inner product. Even if the simulated vectors are made fully orthogonal (inner products equal zero), i.e., $X_t^{\text{sim}} = 10u_t$, our results still hold.

⁴²Given that the original feature dimension is 79, we can generate at most 79 mutually orthogonal vectors.

⁴³This procedure mirrors standard memory experiments, in which participants study a list of items, recall a specific item, and then make free recalls (Healey et al., 2019).

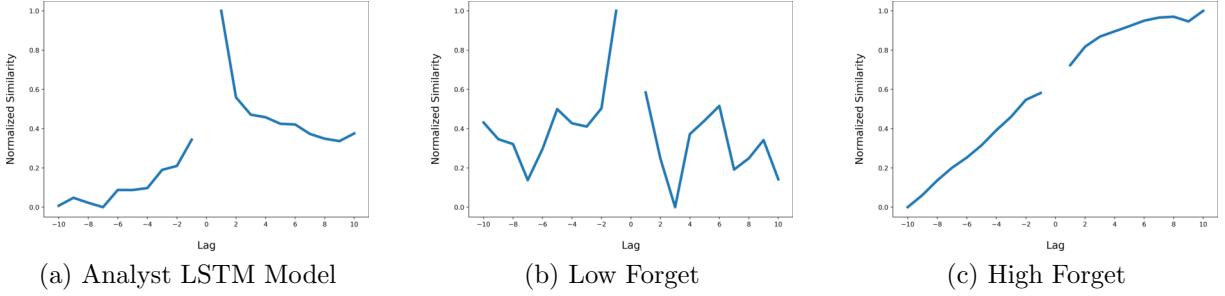


Figure A.1: Temporal Contiguity Effect in the Analyst LSTM Model

This figure plots the average normalized similarity $S(c_{\tau+\ell}^{\text{sim}}, c_{T+1}^{\text{sim}})$ from 10,000 simulations of the estimated analyst LSTM model, and two alternative specifications: one with the forget gate fixed at its 20th percentile (low, forget = 0.079) and one fixed at its 80th percentile (high, forget = 0.937). Inputs are simulated for $T = 70$ periods as $X_t^{\text{sim}} = \mathbb{E}[X] + 10 u_t \times \sigma(X)$, where u_t are mutually orthogonal unit vectors; in period $T + 1$ one input from $\tau \in \{11, \dots, T - 10\}$ is duplicated to cue retrieval. Mental contexts $\{c_t^{\text{sim}}\}$ are produced by the estimated model, and similarity is computed using (2.9). Within each run, similarities are normalized by the minimum and maximum across $\ell \neq 0$. The x -axis reports the lag ℓ relative to τ and the y -axis reports the normalized similarity.

the forget gate to a small number (forget = 0.079, the 20th percentile of its empirical distribution). In this case, the memory cell stores little long-term information, and all content is transitory. Consequently, the analyst almost carries no information forward, and the temporal contiguity pattern disappears: similarity does not decline with $|\ell|$, and similarities at positive lags are not higher than those at negative lags. In Subfigure A.1c, we instead, set the forget gate to a large number (forget = 0.937, the 80th percentile of its empirical distribution), largely removing selectively forgetting (see Section 5.2). The resulting pattern is nearly a rising line, with more recent periods slightly favored. This occurs because the memory cell becomes highly persistent without selective forgetting, clearly violating the temporal-contiguity conditions.

These results show that our analyst LSTM empirically reproduces temporal contiguity, a core principle of human memory. However, the effect is not guaranteed by the structure: under certain parameter settings, the temporal contiguity effect can vanish.

Appendix A.4 Feature-Based Recalls

In this appendix, we report recall patterns when retrieval is cued by the external features X rather than the mental context c and refer these recalls as feature-based recalls. We standardize the external features as described in Section 2.3 . For each firm and recall time x , we compute the similarity between the current external features and all past external features using (3.1)

$$S = \exp(-\|X_{j,\tau} - X_{i,t}\|_2), \quad (6.2)$$

and select the five most similar past experiences as feature-based recalls.

Figure A.2 displays the pattern of feature-based recalls. Feature-based recalls exhibit a dominating recency effect: distant episodes are rarely retrieved except during the COVID-19 period. The feature-based recalls are concentrated, clustering around a few specific dates. By contrast, analyst LSTM recalls include more distant episodes and are more dispersed. The analyst tends to retrieve episodes that are close in time (temporal contiguity) and also display additional temporal links such as cyclical and seasonality.

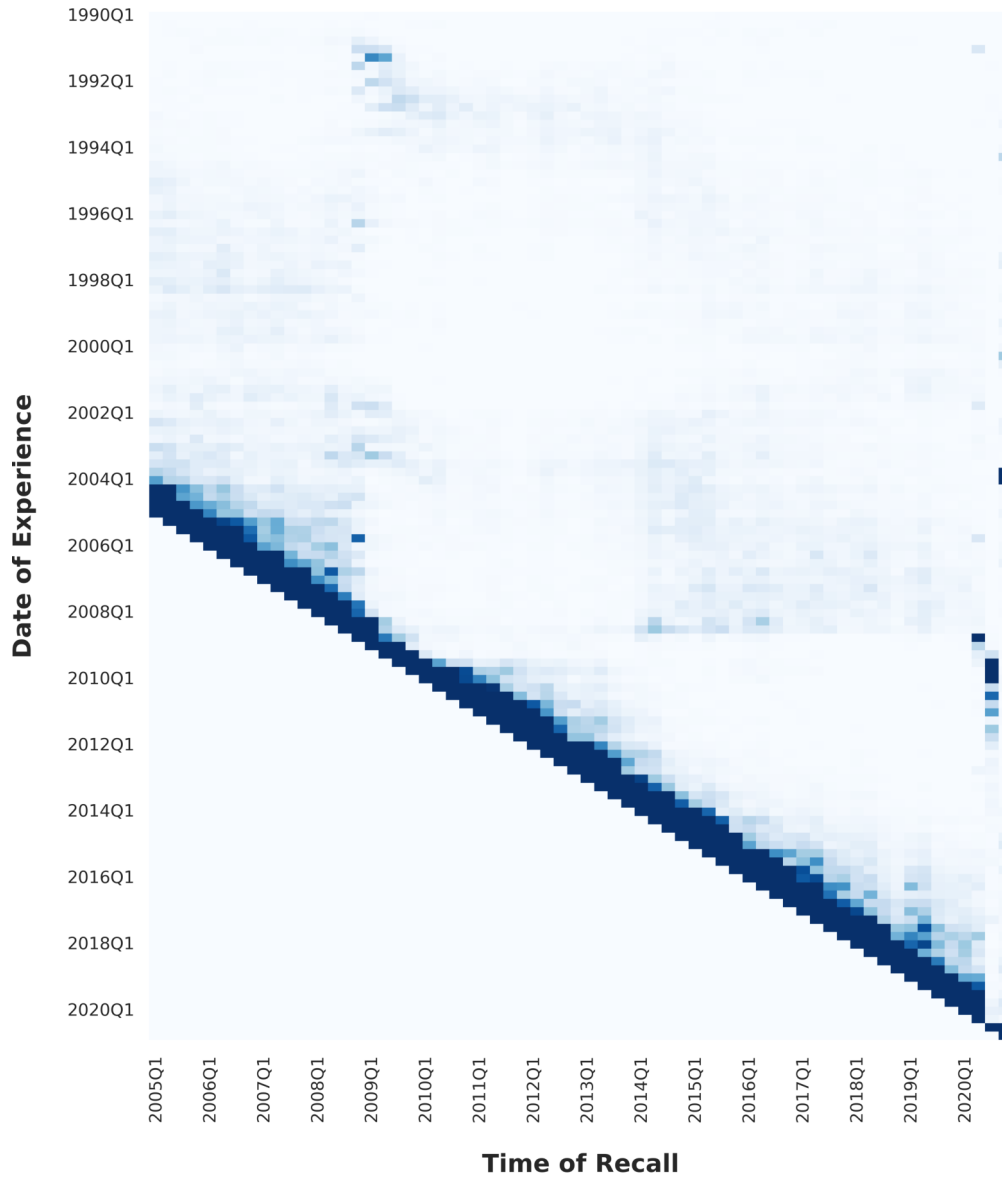


Figure A.2: Feature-Based Recalls

The heatmap shows how often past experiences from date y (rows) are recalled at the current time x (columns) when retrieval is cued by the external features X . For each firm at time x , we compute the similarity between the current external features and all past external features, select the five most similar past experiences, and record their occurrence dates across all firms. The color intensity indicates the percentage of recalls from each date y , with the scale capped at 5% for display clarity. Darker blue areas indicate that experiences from date y are more frequently recalled at time x .

Appendix A.5 Selective Forgetting

This appendix examines the economic impacts of selective forgetting by comparing the predictive power for realized EPS revisions between LSTM models with and without the forget gate.

Table A.3: Realized Earnings Revision and Recall Revisions

	$\Delta RE_{i,t+l}$			
	(1)	(2)	(3)	(4)
Benchmark recalled revisions w/ forget $RR_{i,t}^B$	0.284*** (0.081)			
Benchmark recalled revisions w/o forget $RR_{i,t}^{B/F}$		0.319*** (0.065)		
Analyst recalled revisions w/ forget $RR_{i,t}^A$			0.110*** (0.017)	
Analyst recalled revisions w/o forget $RR_{i,t}^{A/F}$				0.203*** (0.024)
Firm fixed effect	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes
Observations	277,410	277,410	277,410	277,410
R-squared	0.239	0.233	0.262	0.267
Within R-squared	0.019	0.011	0.011	0.019

This table presents results for regressions of the form

$$\Delta RE_{i,t+l} = \beta \times RR_{i,t} + \theta \times Z_{i,t} + \gamma_i + \eta_t \varepsilon_{i,t}.$$

$\Delta RE_{i,t+l}$ is firm i 's realized EPS revision which is available in month $t + l$, i.e., $\Delta RE_{i,t+l} = \frac{EPS_{i,t+l} - F_{i,t-1}[EPS_{i,t+l}]}{P_{i,t-1}}$. $RR_{i,t}$ denotes the recalled revisions, as defined in (3.6). We consider four versions of $RR_{i,t}$: $RR_{i,t}^B$, $RR_{i,t}^{B/F}$, $RR_{i,t}^A$, and $RR_{i,t}^{A/F}$. $RR_{i,t}^B$ is the benchmark LSTM recalled revisions, $RR_{i,t}^{B/F}$ is the benchmark recalled revisions from the counterfactual LSTM without the forget gate, and $RR_{i,t}^A$ and $RR_{i,t}^{A/F}$ are the corresponding recalled revisions for the analyst. The sample period is January 2007 to December 2020. Standard errors are two-way clustered by industry and year and are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

We first assess the predictive power of benchmark memory models with and without the forget gate (Table A.3). Columns (1) and (2) show that recalled revisions from the benchmark model with the forget gate predict realized EPS revisions better than the counterfactual model without it, with within R^2 declining by 40% when selective forgetting is removed. This indicates that the benchmark uses the forget gate effectively to discard less relevant episodes.

The pattern differs for the analyst. Columns (3) and (4) show that recalled revisions without selective forgetting predict realized EPS better than those with it, with within R^2

rising by 68%. This suggests that selective forgetting, while designed to improve memory efficiency, is applied suboptimally by the analyst.

Memory capacity is limited, and both models experience passive forgetting as new information displaces older content. Selective forgetting aims to allocate capacity efficiently by discarding low-value signals. For the benchmark, this mechanism enhances recall informativeness. For the analyst, however, selective forgetting shifts recalls away from the benchmark, reducing predictive power. The distortion stems not from selective forgetting itself, but from how it is applied. Overall within R^2 levels are modest, so these results are suggestive.

Appendix A.6 Robustness Checks

In the main analyses, we limit the memory database size $M_{i,t}$, denoted as N , to the five most similar past experiences. In this appendix, we provide robustness checks for different choices of N as well as additional necessary robustness checks.

Table A.4 shows that recalled revisions positively predict forecast revisions without fixed effects and across different choices of N .

Table A.5 illustrates that the dispersion of recalled revisions positively predicts analyst disagreement and abnormal trading volume, excluding crisis periods and across different choices of N .

Table A.6 confirms that recall distortion positively predicts analyst forecast errors using different choices of N .

Table A.7 tests the robustness of Table VIII using alternative measures of misreactions and recency effects, while Table A.8 provides robustness checks for Table IX with alternative measures of recency effects and choice of N for computing recalled revisions.

Table A.9 demonstrates that the returns on the long-short strategy based on analyst recalled revisions RR^A cannot be explained by leading asset pricing models.

Table A.10 shows the performance of the standard short-term reversal strategy in our sample following Jegadeesh (1990). We sort stocks into terciles on their prior-month returns, then long stocks in the first tercile group (past losers) and short stocks in the third tercile group (past winners).

Table A.11 reports that the returns on the long-short strategy based on analyst recalled revisions RR^A remain effective across different sample periods and choices of N .

Table A.12 presents the returns of double-sorted strategies on biased recall of the prior month and prior-month returns, verifying its validity across different sample periods and choices of N .

Figure A.3 shows that the cumulative returns on the long-short portfolio sorted by recall distortion are robust across varying sample periods and choices of N .

Figure A.4 reports cumulative returns on the double-sorted portfolio based on biased recall of the prior month and prior-month returns, consistent across different sample periods and choices of N .

Figure A.5 displays analyst recalls and their comparison to benchmark recalls across different values of N .

Overall, these robustness checks confirm that our main results hold consistently across varying values of N , time periods, model specifications, and measures.

Table A.4: Analyst Forecast Revisions and Recalled Revisions, across Specifications and Recall Counts

	Forecast Revisions ($\Delta F_{i,t}$)					
	Without Fixed Effects			Different Number of Recalls		
	(1)	(2)	(3)	N=1 (4)	N=3 (5)	N=10 (6)
LSTM recalled revision RR^A	0.078*** (0.019)		0.073*** (0.014)	0.017*** (0.005)	0.043*** (0.010)	0.093*** (0.014)
Naive recalled revision RR^N		0.028* (0.016)	0.007 (0.010)	0.012 (0.008)	0.004 (0.010)	-0.003 (0.012)
Lagged forecast revision			0.154*** (0.051)	0.132*** (0.042)	0.130*** (0.043)	0.121** (0.043)
Lagged earnings growth			0.011** (0.005)	0.013** (0.005)	0.013** (0.005)	0.013** (0.005)
Lagged forecast			-2.328*** (0.615)	-4.611*** (0.746)	-4.648*** (0.750)	-4.730*** (0.772)
Firm fixed effect	No	No	No	Yes	Yes	Yes
Month fixed effects	No	No	No	Yes	Yes	Yes
Observations	277,487	277,487	277,487	277,410	277,410	277,410
R-squared	0.012	0.002	0.025	0.067	0.069	0.073
Within R-squared	0.012	0.002	0.025	0.021	0.023	0.028

This table presents the robustness check results for Table II without fixed effects in Columns (1)-(3) and using different number of the most similar past experiences (N) in memory database M in Columns (4)-(6). The regressions follow the form

$$\Delta F_{i,t} = \beta \times RR_{i,t} + \theta \times Z_{i,t} + \gamma_i + \eta_t + \varepsilon_{i,t},$$

where $\Delta F_{i,t}$ is the consensus analyst forecast revision (defined in (3.5)) and $RR_{i,t}$ is the recalled revision. We consider two measures of $RR_{i,t}$: $RR_{i,t}^A$, the recalled revision based on the analyst LSTM model (defined in (3.6)), and $RR_{i,t}^X$, the recalled revision based on external features (defined in (3.7)). $Z_{i,t}$ is a vector of control variables, and γ_i and η_t denote firm and time fixed effects, respectively. Columns (1) and (2) are specifications without controls. Columns (3)-(6) add three controls: (i) lagged forecast revision ($F_{i,t-1}[EPS_{i,t+l}] - F_{i,t-2}[EPS_{i,t+l}]/P_{i,t-1}$); (ii) lagged earnings growth ($EPS_{i,t-l_1} - EPS_{i,t-l_2}$)/ $P_{i,t-1}$; and (iii) the lagged forecast $F_{i,t-1}[EPS_{i,t+l}]/P_{i,t-1}$, where $F_{i,t}[EPS_{i,t+l}]$ is the consensus analyst forecast for firm i 's future EPS at time t , $EPS_{i,t-l_1}$ and $EPS_{i,t-l_2}$ are realized earnings over the prior two fiscal years, and $P_{i,t-1}$ is the lagged stock price. The sample period is January 2007 to December 2020. Standard errors are two-way clustered by industry and year and are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table A.5: Recall dispersion, Disagreement, and Trading volume, across Sample Periods and Recall Counts

	Disagreement _{<i>i,t</i>}			Abnormal volume AbVol _{<i>i,t</i>}		
	Without Crisis (1)	N=3 (2)	N=10 (3)	Without Crisis (4)	N=3 (5)	N=10 (6)
$\sigma(rr)_{i,t}$	0.005* (0.003)	0.008*** (0.002)	0.009*** (0.002)	0.057*** (0.018)	0.036** (0.013)	0.046*** (0.012)
$\sigma(rr)_{i,t-1}$	0.005** (0.002)	0.009*** (0.002)	0.006** (0.002)	-0.024 (0.021)	0.003 (0.019)	0.010 (0.013)
Disagreement _{<i>i,t-1</i>}	0.474*** (0.019)	0.540*** (0.017)	0.539*** (0.017)			
AbVol _{<i>i,t-1</i>}				0.521*** (0.008)	0.539*** (0.010)	0.539*** (0.010)
$Ret_{i,t-1}$	-0.009*** (0.001)	-0.012*** (0.002)	-0.012*** (0.002)	-0.096** (0.040)	-0.086** (0.035)	-0.086** (0.035)
$Ret_{i,t-2}$	-0.003*** (0.001)	-0.004*** (0.001)	-0.003*** (0.001)	-0.003 (0.020)	0.004 (0.019)	0.005 (0.020)
$Ret_{i,t-3}$	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	0.043 (0.024)	0.055** (0.025)	0.055** (0.025)
$\sigma(Ret)_{i,t-1}$	0.037*** (0.005)	0.039*** (0.006)	0.038*** (0.006)	-0.707*** (0.107)	-0.574*** (0.110)	-0.576*** (0.110)
Firm fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	46,461	59,160	59,160	158,424	194,444	194,444
R-squared	0.593	0.608	0.609	0.384	0.412	0.412

This table presents the robustness check results for Table IV with different sample periods and using different number of the most similar past experiences (N) in memory database M . Columns (1) to (3) present the regression results on disagreement and columns (4) to (6) present the regression results on trading volume. The sample period is January 2007 to December 2020, while Columns (1) and (4) present the regression results excluding the crisis periods, specifically the years 2007, 2008, and 2020. Columns (2) and (5) present the regression results using three most similar past experiences ($N = 3$). Columns (3) and (6) present the regression results using ten most similar past experiences ($N = 10$). The regressions follow the form

$$\text{Disagreement}_{i,t} = \beta \times \sigma(rr)_{i,t} + \theta \times Z_{i,t} + \gamma_i + \eta_t + \varepsilon_{i,t},$$

and

$$\text{AbVol}_{i,t} = \beta \times \sigma(rr)_{i,t} + \theta \times Z_{i,t} + \gamma_i + \eta_t + \varepsilon_{i,t},$$

Recall dispersion $\sigma(rr)_{i,t}$ is the weighted standard deviation of recalled revisions for firm i in month t , defined in (3.10). Disagreement_{*i,t*} is the cross-sectional standard deviation of individual analyst EPS forecasts scaled by last month's stock price, following Diether et al. (2002). AbVol_{*i,t*} is abnormal log trading volume for firm i in month t , following Cookson and Niessner (2020), which equals log volume in month t minus the average log volume from months $t-12$ to $t-2$. The control variables $Z_{i,t}$ includes lagged monthly returns $Ret_{i,t-1}$, $Ret_{i,t-2}$, and $Ret_{i,t-3}$, and return volatility $\sigma(Ret)_{i,t-1}$, measured as the standard deviation of monthly returns over months $t-12$ through $t-1$. All specifications include firm and time fixed effects γ_i and η_t . Coefficients on recall dispersion $\sigma(rr)_{i,t}$ are scaled by 1,000 for readability. Standard errors are two-way clustered by industry and year and are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table A.6: Analyst Forecast Errors and Recall Distortion across Recall Counts

	Forecast error $e_{i,t}$		Forecast error $ e_{i,t} $		Forecast direction error $e_{i,t}^d$	
	N=3	N=10	N=3	N=10	N=3	N=10
	(1) Linear	(2) Linear	(3) Linear	(4) Linear	(5) Ologit	(6) Ologit
$\Delta R_{i,t}$	0.033*** (0.008)	0.038*** (0.007)			0.018*** (0.003)	0.027*** (0.004)
$ \Delta R_{i,t} $			0.095*** (0.023)	0.095*** (0.019)		
Lagged forecast revision	-0.008 (0.019)	-0.007 (0.019)	-0.032** (0.014)	-0.032** (0.014)	0.020** (0.008)	0.020** (0.009)
Lagged earnings growth	0.004 (0.002)	0.004 (0.002)	-0.010** (0.003)	-0.010** (0.003)	0.002 (0.001)	0.002 (0.001)
Lagged forecast	1.252** (0.499)	1.251** (0.502)	-2.679*** (0.540)	-2.677*** (0.552)	0.390* (0.225)	0.383* (0.226)
Firm fixed effect	Yes	Yes	Yes	Yes	No	No
Month fixed effects	Yes	Yes	Yes	Yes	No	No
Observations	277,410	277,410	277,410	277,410	277,487	277,487
R-squared	0.145	0.145	0.302	0.300		

This table presents the robustness check results for Table V using different number of the most similar past experiences (N) in memory database M to form recall distortions ($\Delta R_{i,t}$). Columns (1), (3), and (5) present the regression results using three most similar past experiences ($N = 3$). Columns (2), (4), and (6) present the regression results using ten most similar past experiences ($N = 10$). Columns (1)–(2) estimate

$$e_{i,t} = \beta \times \Delta R_{i,t} + \theta \times Z_{i,t} + \gamma_i + \eta_t + \varepsilon_{i,t}.$$

Columns (3)–(4) estimate

$$|e_{i,t}| = \beta \times |\Delta R_{i,t}| + \theta \times Z_{i,t} + \gamma_i + \eta_t + \varepsilon_{i,t}.$$

Columns (5)–(6) estimate an ordered logit for the direction of the forecast error,

$$\Pr(e_{i,t}^d = j) = \Lambda(\kappa_j - x_{i,t}) - \Lambda(\kappa_{j-1} - x_{i,t}), \quad j \in \{-2, -1, 0, 1, 2\},$$

where $x_{i,t} = \beta \times \Delta R_{i,t} + \theta \times Z_{i,t} + \gamma_i + \eta_t$ and $\Lambda(z) = 1/(1 + \exp(-z))$. The cutoffs κ_j are estimated jointly with the coefficients. The dependent variables are the analyst forecast error, $e_{i,t}$, the magnitude of analyst forecast error $|e_{i,t}|$, and the error direction $e_{i,t}^d \in \{-2, -1, 0, 1, 2\}$, for firm i at time t . $\Delta R_{i,t}$ is the analyst recall distortion defined in (4.1). All columns add three controls: (i) lagged forecast revision $(F_{i,t-1}[EPS_{i,t+l}] - F_{i,t-2}[EPS_{i,t+l}])/P_{i,t-1}$; (ii) lagged earnings growth $(EPS_{i,t-l_1} - EPS_{i,t-l_2})/P_{i,t-1}$; and (iii) the lagged forecast $F_{i,t-1}[EPS_{i,t+l}]/P_{i,t-1}$, where $F_{i,t}[EPS_{i,t+l}]$ is the consensus analyst forecast for firm i 's future EPS at time t , $EPS_{i,t-l_1}$ and $EPS_{i,t-l_2}$ are realized earnings over the prior two fiscal years, and $P_{i,t-1}$ is the lagged stock price. The sample period is January 2007 to December 2020. Standard errors are two-way clustered by industry and year and are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table A.7: Misreactions in Perceptions, Forecast Misreactions, and Biased Recency Effect, across Sample Periods and Measures

	Misreaction		Biased Recency Effect			
	ΔReact_t incl.	ΔReact_t	24 months		36 months	
	$\text{sign}(\Delta F_{i,t}) \neq \text{sign}(\Delta RE_{i,t})$	excl. $\Delta F_{i,t} = 0$	Normal	Crisis	Normal	Crisis
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta^2\omega$	-0.022*** (0.006)	-0.020*** (0.006)	0.021** (0.010)	-0.019*** (0.004)	0.020* (0.010)	-0.019*** (0.004)
Observations	180	180	144	36	144	36
R-squared	0.099	0.163	0.033	0.251	0.024	0.295

This table presents the robustness check results for Table VIII with alternative measures of misreactions and biased recency effects. The independent variable, $\Delta^2\omega_t$, is defined in Equation (5.1). The dependent variables, misreaction ΔReact_t and biased recency effect $\Delta\text{Recency}_t$ are similar to the definition in Equations (5.3), (5.5), respectively. We examine two variations of ΔReact_t : including the cases when $\text{sign}(\Delta F_{i,t}) \neq \text{sign}(\Delta RE_{i,t})$ in Column (1), and excluding the cases when $\Delta F_{i,t} = 0$ in Column (2). Columns (3) and (4) measure recency effects by analyst recalls falling within the last 24 months. Columns (5) and (6) measure recency effects by analyst recalls falling within the last 36 months. The sample period is January 2007 to December 2020, with 2008, 2009, and 2020 are classified as crisis periods, while the remaining years are considered normal periods. Standard errors are two-way clustered by industry and year and are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table A.8: Paired t -statistics Between LSTM Models with and without Selective Forgetting, across Measures and Recall Counts

Model	Recency Effect		Recalled Revisions		
	24 months (1)	36 months (2)	N=1 (3)	N=3 (4)	N=10 (5)
<i>Panel A: Analysts</i>					
LSTM with forget gate	0.292	0.360	-0.687	-0.706	-0.708
LSTM without forget gate	0.446	0.520	-0.548	-0.604	-0.647
Difference	0.153	0.160	0.139	0.102	0.061
t -statistics	(8.66)	(9.28)	(8.36)	(7.56)	(5.14)
<i>Panel B: Benchmark</i>					
LSTM with forget gate	0.585	0.636	-0.330	-0.403	-0.467
LSTM without forget gate	0.570	0.620	-0.412	-0.460	-0.515
Difference	-0.015	-0.017	-0.081	-0.057	-0.048
t -statistics	(-1.72)	(-1.81)	(-5.56)	(-3.91)	(-4.06)

This table presents the robustness check results for Table IX with alternative measures of recency effects and recalled revisions. Columns (1) and (2) present the paired t -tests of recency effects (Recency, defined in (5.4)) for 24 months and 36 months. Columns (3) to (5) present the t -tests of recalled revisions (RR , defined in (3.6)) using different numbers of the most similar past experiences (N). Panel A reports results for the full analyst LSTM versus a counterfactual without the forget gate. Panel B reports the corresponding results for the benchmark LSTM. The sample period is January 2007 to December 2020. The t -statistics reported in parentheses are robust to heteroskedasticity and autocorrelation with up to 12 lags.

Table A.9: Performance of Portfolios Sorted by Analyst Recalled Revisions

	CAPM		FF3		Carhart	
	Coef	<i>t</i> -stat	Coef	<i>t</i> -stat	Coef	<i>t</i> -stat
Intercept	0.55	2.97	0.41	2.55	0.41	3.01
MKT- R_f	-0.16	-2.56	-0.10	-1.89	-0.05	-0.84
SMB			-0.04	-0.36	-0.01	-0.11
HML			-0.24	-3.28	-0.10	-1.88
MOM					0.21	2.89

This table reports regressions of the long-short portfolio returns, formed by sorting on the analyst recalled revision RR^A (defined in Eq. (3.6)). Risk-adjusted returns are reported in percent and are estimated using the CAPM model, the Fama-French three-factor model (FF3) (Fama and French, 1993), and the Carhart four-factor model Carhart (1997). The sample period is January 2007 to December 2020. The *t*-statistics are robust to heteroskedasticity and autocorrelation with up to 12 lags.

Table A.10: Performance of the Standard Short-term Reversal Strategy

	Excess Return		CAPM		FF3		Carhart	
	Coef	<i>t</i> -stat	Coef	<i>t</i> -stat	Coef	<i>t</i> -stat	Coef	<i>t</i> -stat
Intercept	0.48	4.12	0.29	2.09	0.34	2.16	0.34	2.16
MKT- R_f			0.25	4.11	0.22	3.69	0.22	3.70
SMB					0.07	0.84	0.07	0.78
HML					0.07	0.64	0.07	0.82
MOM							-0.01	-0.13

This table presents the equal-weighted raw returns and risk-adjusted returns of the standard short-term reversal trading strategy. The strategy sorts stocks into terciles according to prior-month returns, then long past losers and short past winners. All returns are reported in percent and risk-adjusted return are estimated using the CAPM model, the Fama-French three-factor model (FF3) (Fama and French, 1993), and the Carhart four-factor model Carhart (1997). The sample period is January 2007 to December 2020. The *t*-statistics are robust to heteroskedasticity and autocorrelation with up to 12 lags.

Table A.11: Recalled Revisions, Forecast Revisions, and Stock Returns, across Sample Periods and Recall Counts

	CAPM	FF3	Carhart
<i>Panel A: Different sample periods</i>			
2007 - 2019	0.48 (2.91)	0.38 (2.37)	0.36 (3.09)
2010 - 2019	0.49 (2.81)	0.44 (2.81)	0.35 (2.09)
2010 - 2020	0.58 (3.65)	0.50 (3.30)	0.47 (2.76)
<i>Panel B: Different number of past experiences in M</i>			
$N = 1$	0.42 (3.10)	0.29 (2.03)	0.29 (2.54)
$N = 3$	0.55 (3.35)	0.42 (2.67)	0.42 (3.44)
$N = 10$	0.49 (2.44)	0.36 (1.94)	0.37 (2.79)

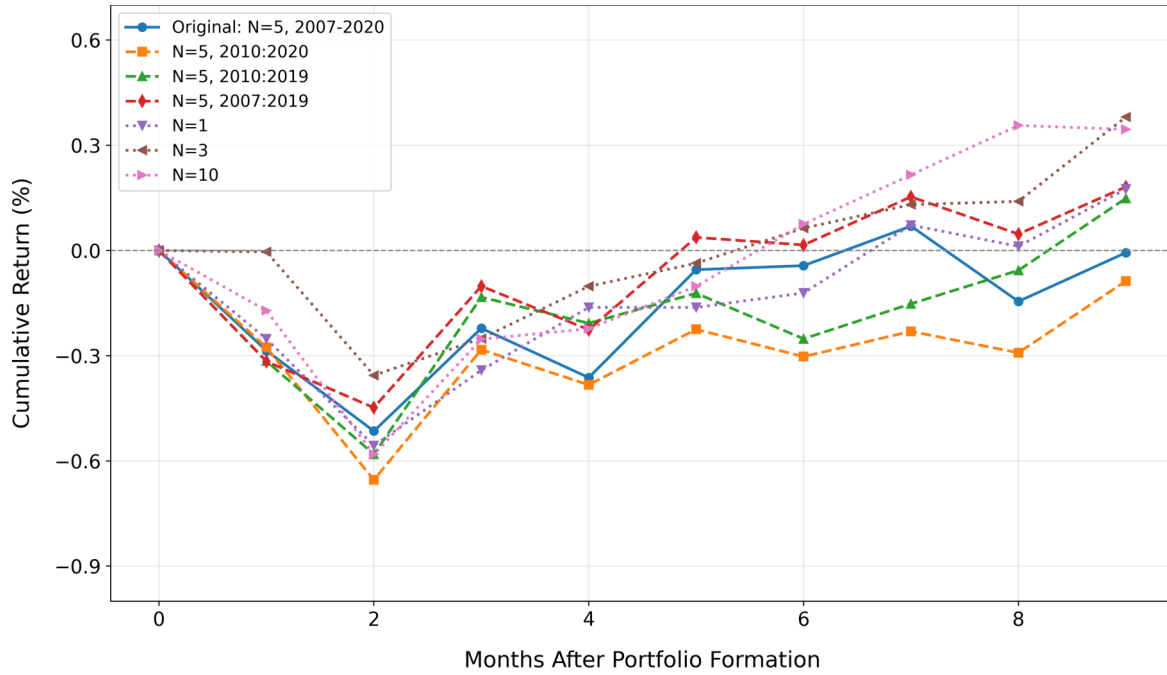
This table reports the robustness check for Table III in different sample periods (Panel A) and using different number of the most similar past experiences (N) in memory database M to form the analyst recalled revisions (RR^A) (Panel B). This table presents average risk-adjusted returns for value-weighted long-short portfolios sorted by based on the analyst recalled revision (RR^A , defined in (3.6)). Risk-adjusted returns are reported in percent and are estimated using the Carhart four-factor model Carhart (1997). For Panel B, the sample period is January 2007 to December 2020. The t -statistics reported in parentheses are robust to heteroskedasticity and autocorrelation with up to 12 lags.

Table A.12: Biased Recall of the Prior Month and Short-Term Reversal, across Sample Periods and Recall Counts

$\Delta\rho$	1	2	3	3 – 1
<i>Panel A: Different sample periods</i>				
2007 - 2019	-0.59 (-1.74)	0.36 (2.69)	0.55 (2.38)	1.14 (3.78)
2010 - 2019	-0.51 (-1.35)	0.35 (2.22)	0.60 (2.93)	1.11 (2.88)
2010 - 2020	-0.53 (-1.52)	0.32 (2.06)	0.53 (2.51)	1.07 (2.97)
<i>Panel B: Different number of past experiences in M</i>				
$N = 3$	-0.53 (-1.13)	0.41 (2.64)	0.40 (1.74)	0.93 (2.16)
$N = 10$	-0.35 (-1.12)	0.38 (2.65)	0.43 (1.80)	0.77 (2.39)

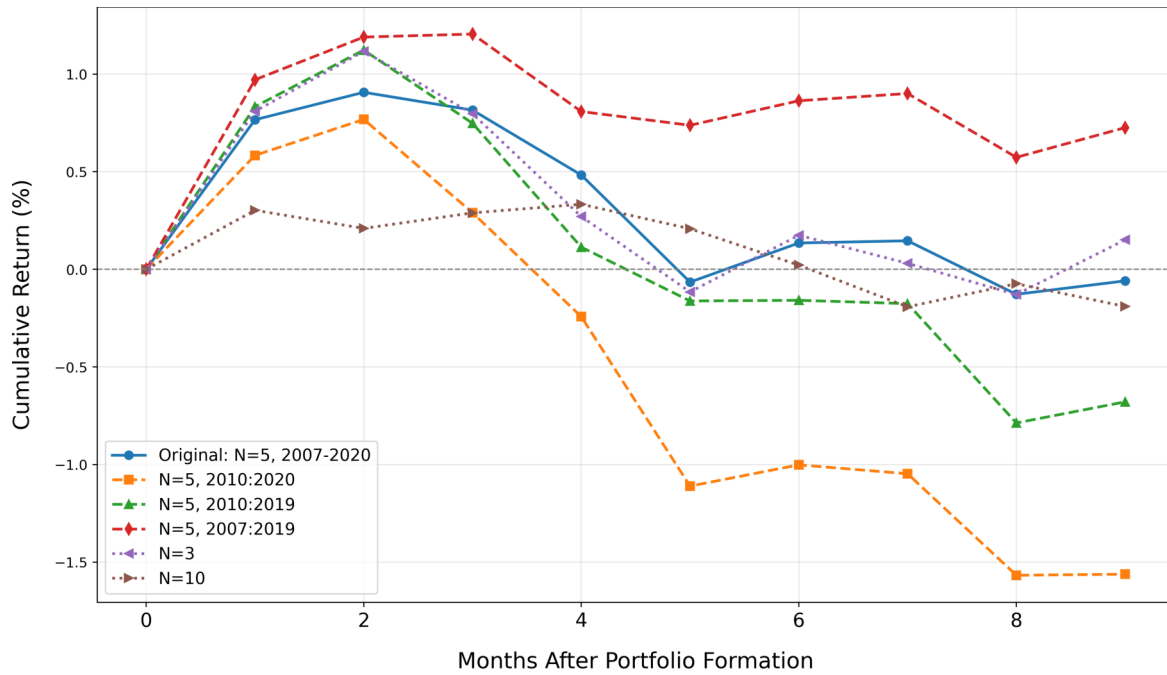
This table reports the robustness check for Table VII in different sample periods (Panel A) and using different number of the most similar past experiences (N) in memory database M to form the biased recall of the prior month $\Delta\rho$ (Panel B). We report average risk-adjusted returns (Carhart, 1997) for short-term reversal strategies in different groups of biased recall ($\Delta\rho$, defined in (4.6)). For Panel B, the sample period is January 2007 to December 2020. The t -statistics reported in parentheses are robust to heteroskedasticity and autocorrelation with up to 12 lags.

Figure A.3: Cumulative Returns on Long-Short Portfolio Sorted by Recall Distortion, across Sample Periods and Recall Counts



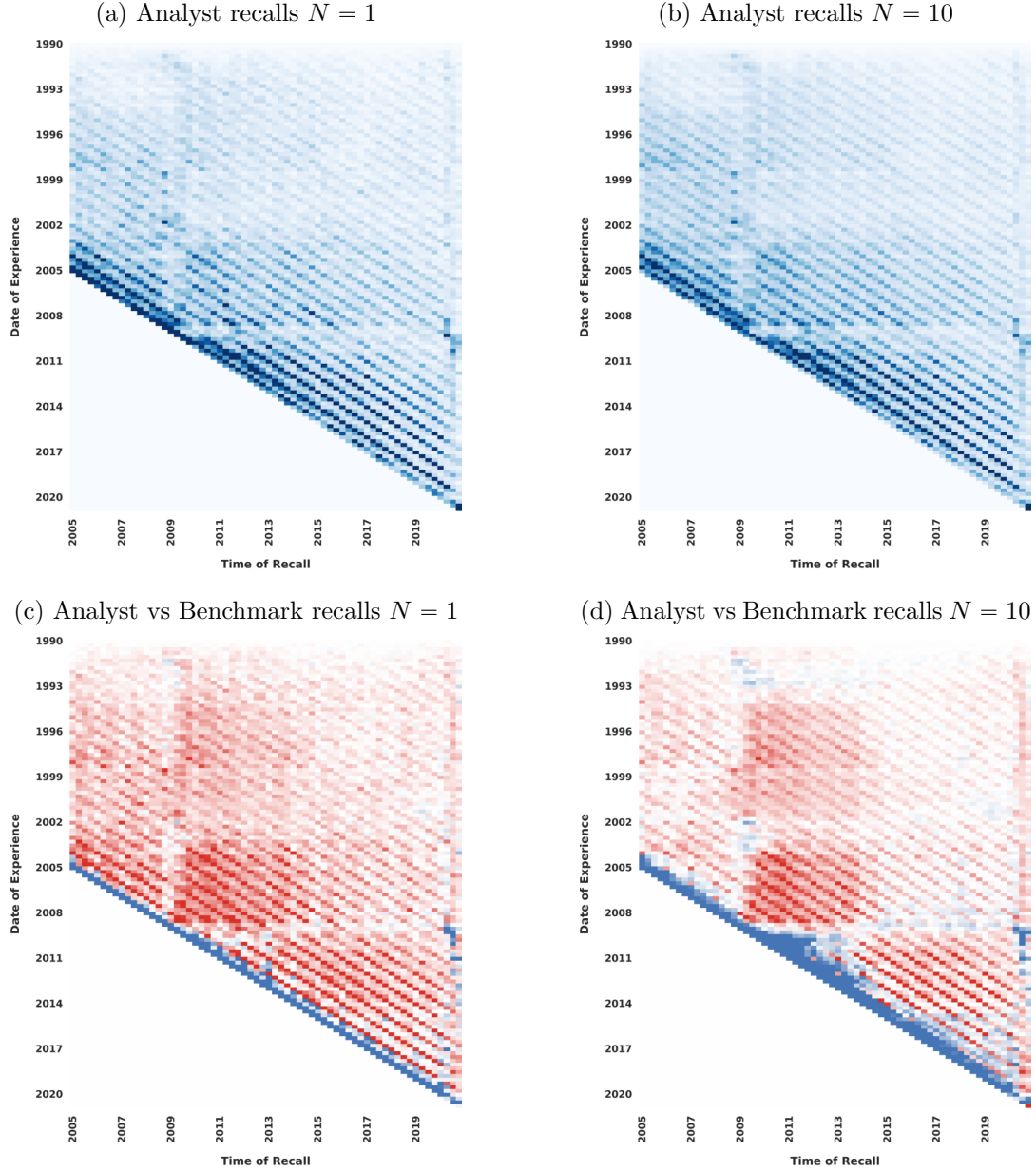
This figure reports robustness check for Figure VII by varying the sample periods and the number of most similar past experiences (N) in the memory database M . Each line plots the cumulative returns of a value-weighted long-short strategy sorted by recall distortion (ΔR , defined in (4.1)). The portfolio goes long the quintile of stocks with the most negative recall distortion and shorts the quintile with the most positive recall distortion. The orange, green, and red lines correspond to sample periods 2010–2020, 2010–2019, and 2007–2019, respectively, holding $N = 5$. The purple, brown, and pink lines correspond to $N = 1$, $N = 3$, and $N = 10$, respectively, using the full 2007–2020 sample.

Figure A.4: Cumulative Returns of Double-Sorted Long-Short Portfolio by Biased Recall of the Prior Month and Short-Term Reversal, across Sample Periods and Recall Counts



This figure reports robustness check for Figure VIII by varying the sample periods and the number of most similar past experiences (N) in the memory database M . Each line plots the cumulative returns of a monthly long-short strategy combining biased recall of the prior month with short-term reversal. Each month, stocks are first sorted into terciles by biased recall ($\Delta\rho$ in (4.6)). Within each $\Delta\rho$ tercile, firms are sorted by prior-month returns, and a reversal portfolio is formed as past losers minus past winners. The strategy goes long the reversal portfolio in the under-recall tercile and shorts it in the over-recall tercile. The orange, green, and red lines correspond to sample periods 2010–2020, 2010–2019, and 2007–2019, respectively, holding $N = 5$. The purple and brown lines correspond to $N = 3$ and $N = 10$, respectively, using the full 2007–2020 sample.

Figure A.5: Analyst LSTM Recalls and Difference from Benchmark LSTM Recalls across Recall Counts



This figure reports robustness check for Figure III and Figure VI by varying the number of most similar past experiences (N) in the memory database M . The heatmap shows how often past experiences from date y (rows) are recalled at the current time x (columns). For each model, firm, and recall date x , we compute the similarity between the current mental context and all past mental contexts estimated by that model, select the single ($N = 1$) or the ten ($N = 10$) most similar past experiences, and count their occurrence dates across firms. The color intensity indicates the percentage or the difference in percentage of recalls from each date y . In subfigures A.5a and A.5b, darker blue areas indicate that experiences from date y are more frequently recalled at time x , with a scale capped at 5% for display clarity. In subfigures A.5c and A.5d, darker red indicates dates that the analyst recall more often than the benchmark, while darker blue indicates dates that the benchmark recalls more often than the analyst, with a scale capped at $\pm 2.5\%$ for display clarity.